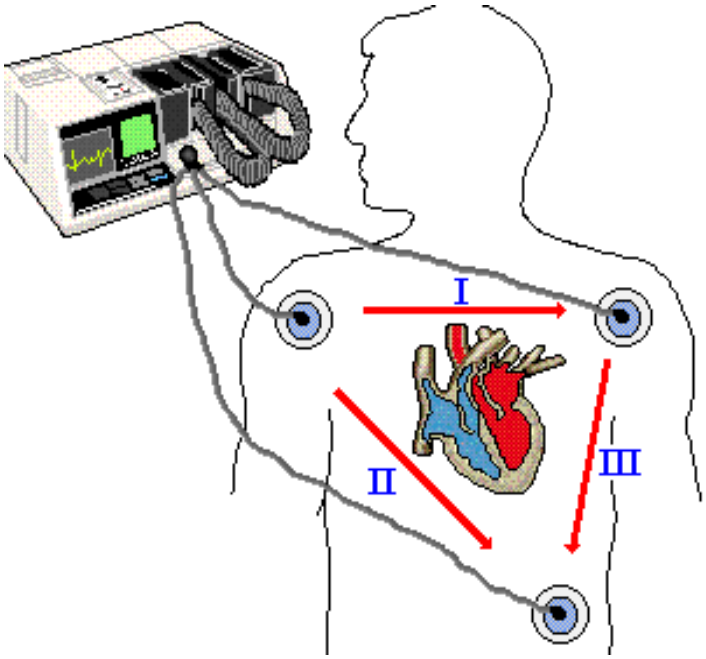
INTRODUCTION

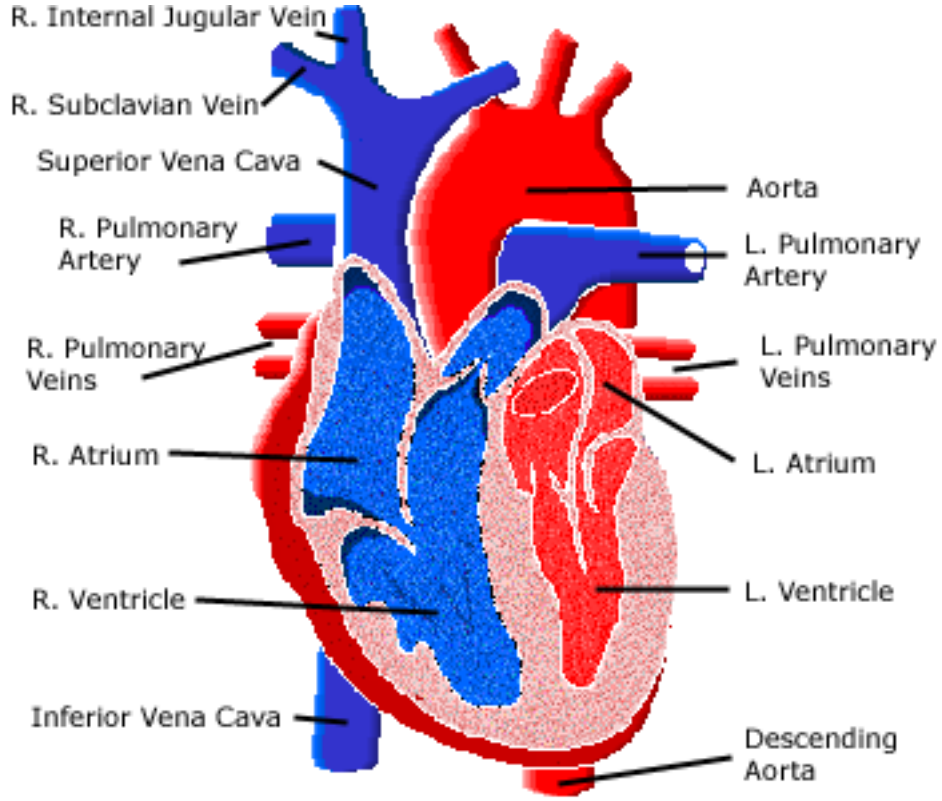


An electrocardiogram, or ECG, is a recording of the electrical activity of the heart. The ECG machine is connected to a person through three leads, or wires that are placed onto the skin of the chest and abdomen. Through these leads, the ECG machine is able to detect the levels of electrical activity within the heart and trace a representation of them onto paper.

The leads are wires that are attached to the patient with adhesive patches. The patches have a gel on their surface that makes contact with the skin and helps the machine detect the electrical signals more clearly. The simplest and most common form of ECG is called the three-lead ECG. We won't go into detail about different lead views yet, but it's important to know that each lead, like a camera, reveals a different perspective depending on what side and angle at which it is focused. The more leads, or perspectives you examine, the more you know about the entire heart. Only one lead is needed to monitor the basic electrical function of the heart. The most common lead used is Lead Two. You can tell from the picture, that Lead Two travels from the top of the heart to the bottom left portion of the heart. This provides a good, overall picture of the heart's conduction system.

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PULMONARY CIRCULATION

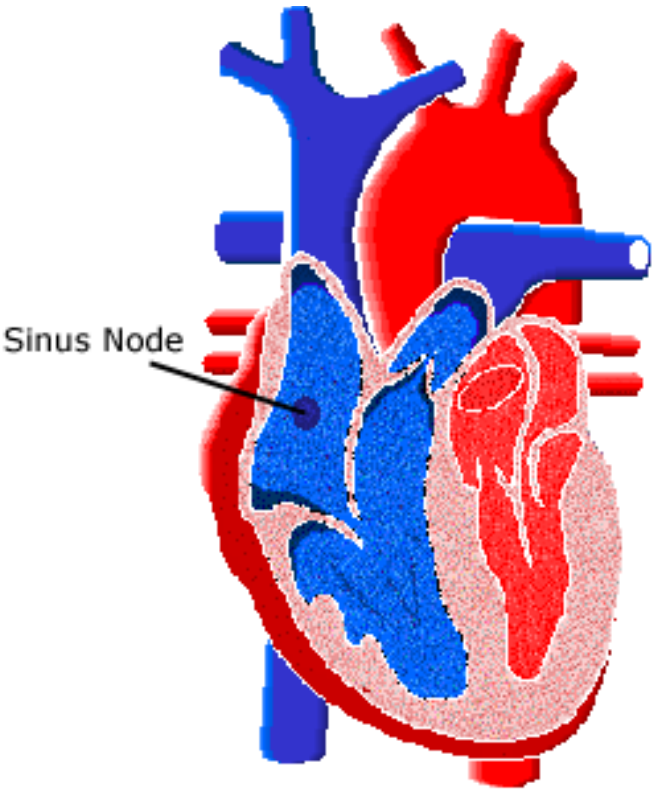


Pulmonary circulation is the movement of blood from the heart, to the lungs, and back to the heart again. This is just one phase of the overall circulatory system.

Blood from the peripheral tissues flows back toward the heart through small veins which eventually merge into the superior and inferior vena cava. These large veins empty the waste-rich blood into the right atrium. As the atria contract, the blood is emptied into the right ventricle. The ventricles contract forcefully and the blood is sent through the pulmonary arteries to the lungs where it is re-oxygenated. The blood then returns via the pulmonary veins and empties into the left atrium. When the atria contract again, the blood is pumped into the left ventricle, which contracts forcefully to send the blood through the aorta, and out to all the body tissues to deliver oxygen.

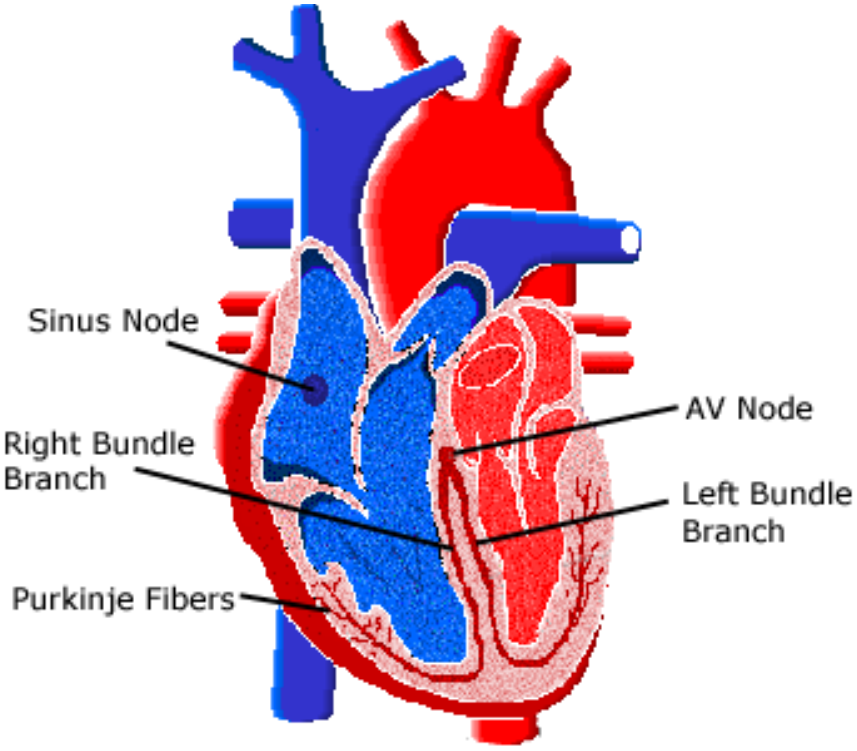
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ELECTROPHYSIOLOGY



The heart, like any machine, has to have an electrical supply for it to work. Almost any group of cells within the heart can act as the electrical supply and originate an impulse to cause the heart to beat. Most often, though, the area that originates the majority of the impulses is the area in the right atrium called the sinus node. It may seem odd that one area normally supplies all of the electrical impulses, when so many areas in the heart are capable of it. The distinguishing feature about the sinus node, is that it normally fires the fastest, and the impulse area, or focus, that fires that fastest will dominate. There are, of course, exceptions to this rule. Sometimes, one of these other areas will become irritated and it will fire a spontaneous beat, also known as an ectopic beat, that disrupts the normal rhythm. Any beat that does not originate in the sinus node is known as an ectopic beat. Sometimes, entire rhythms are formed from ectopic and irregular beats. That is exactly what makes interpreting ECG's so interesting, as you will see.

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Normally, the sinus node beats between 60-100 times a minute. Every time it beats, it sends an impulse through the atria that reaches the AV node.

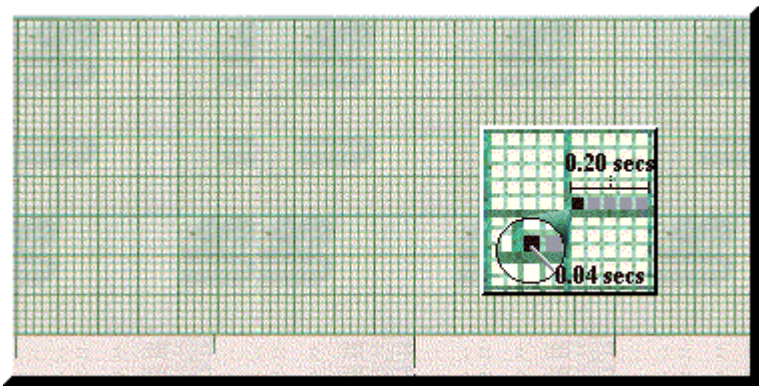
The purpose of the AV node is to delay the impulse just long enough for the atria to contract and send blood into the ventricles. After being delayed in the AV node, the impulse travels down both the left bundle branch and the right bundle branch simultaneously. These branches lead into the left and right ventricles respectively. Then the impulse travels from the bundle branches into the Purkinje fibers. These fibers are tiny strands that deliver the impulse to the muscle cells in the ventricles and cause them to contract.

If for some reason, the sinus node is not generating electrical impulses fast enough or not at all, the AV node acts as a backup and will begin to generate impulses itself. It can beat about 40-60 times a minute. If the AV node fails as well, the body has another backup.

The ventricles can generate impulses on their own at about 20-40 beats a minute. Sometimes different areas of the heart will also generate impulses, and sometimes the AV node and ventricles can generate abnormal rhythms that override the sinus node's ability to function.

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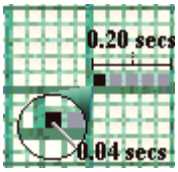
ECG PAPER



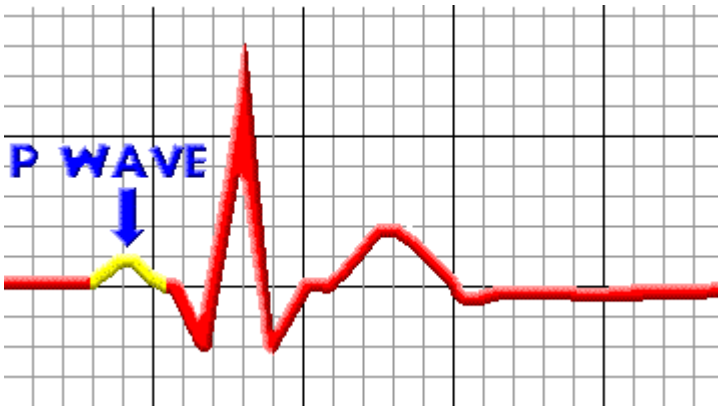
The ECG is recorded on the paper similar to that shown above. The smallest boxes you see are 1 millimeter high and 1 millimeter long. Between each of the dark green lines are five small boxes. The ECG paper records the electrical impulses at a certain speed. The paper moves at 25 millimeters per second while the stylus traces the impulses. This means that each of the smallest boxes represents 0.04 seconds and the space between each of the dark green lines is five times that, or 0.20 seconds.

To help you assess the spacing and duration of different components of the ECG, the small reference grid above will be placed at the top of each page.

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P WAVE

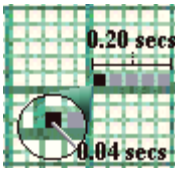


The first segment of a normal ECG is the P wave. This wave is a small, upright deflection that occurs before the QRS complex and signifies atrial depolarization.

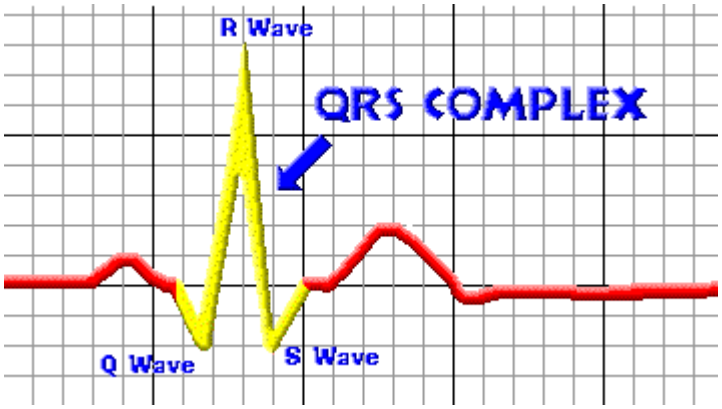
Depolarization refers to the electrical stimulus that passes through the cardiac muscle cells and prepares them for contraction. All rhythms that originate from the sinus node have a P wave. Even ectopic beats that originate in the atria have a P wave, but it may be shaped slightly different from the P waves in the normal beats around it.

The P wave is normally seen before the QRS, as it is in this diagram, but there are a few arrhythmias that do not even have a P wave at all.

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QRS COMPLEX



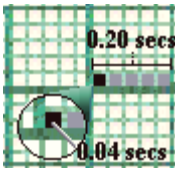
The QRS complex is actually a combination of upward and downward deflections, or waves that, together represent ventricular depolarization (remember that depolarization is the electrical activity that prepares the muscle cell to contract).

The first downward deflection is called a Q wave. The positive deflection is called an R wave and the downward deflection that occurs after the R wave is called the S wave. The confusing aspect of the QRS complex, is that it does not always contain all three waves. What you will notice later in the program, is that quite often, there is no Q wave at all.

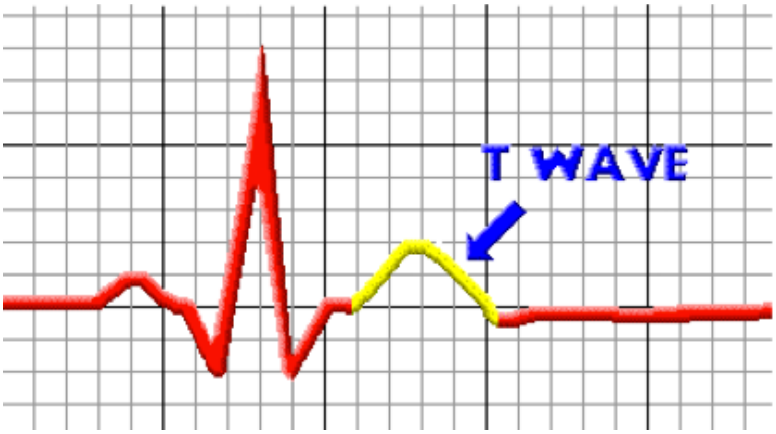
Usually, the first segment of the QRS complex is an upward deflection, or R wave. In fact, in some circumstances, the presence of the Q wave can indicate damage to the heart. Some examples of different QRS complex configurations are shown on the lower left corner of this page. It is important to measure the QRS complex while you are examining it. To do this, you simply count the number of small blocks between the start of the first wave (whether it is a Q wave or an R wave) and the end of the last wave. Normally, the QRS complex is between 1 and 3 small blocks in width (0.04 seconds to 0.12 seconds).



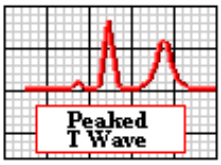
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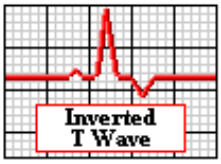


T WAVE



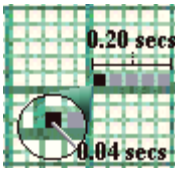
The T wave represents ventricular repolarization. Repolarization just means that the muscle cells in the ventricles are in a resting state. They are resetting themselves so that they can depolarize and contract again when the next electrical impulse passes through them. Although the T wave may resemble the P wave, it can be easily distinguished because it is taller and wider. It also always follows the QRS complex, whereas the P wave precedes the QRS complex.



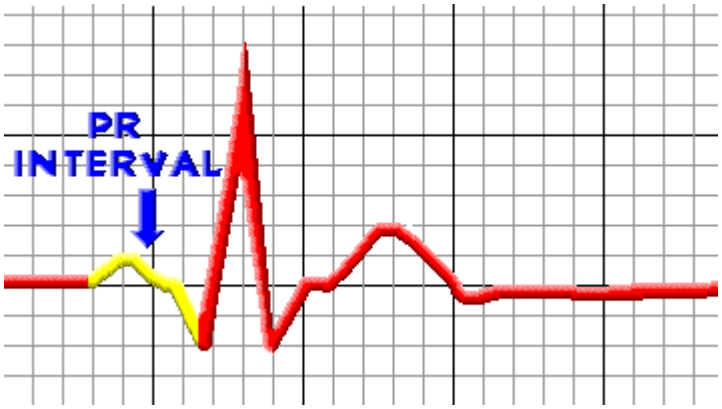


There are two attributes you need to observe about the T wave. First, notice whether or not the T wave is upright or inverted. An inverted T wave may sometimes represent a lack of oxygen to the heart. Also, if the T wave is upright, notice how tall it is. This is referred to as amplitude. Normally, a T wave is no taller than 5 millimeters (5 small boxes). If it is taller than this, it is referred to as a peak T wave. Although it may occur normally in some cases, a peak T wave can be the result of electrolyte abnormalities.

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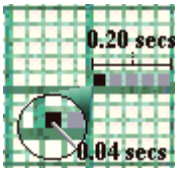
PR INTERVAL





The PR interval represents the amount of time it takes the electrical impulse to travel from it's point of origin in the atria to the AV Node. In other words, it is the time it takes the impulse to get from the sinus node (or an ectopic focus in the atria) to the ventricles. The PR interval is normally from 0.12 seconds to 0.20 seconds (three to five small boxes) and begins at the beginning of the P wave and ends at the beginning of the Q wave, or R wave, if there is no Q wave. For now, remember to look for a PR interval that is longer than FIVE SMALL BOXES. If it is longer than this, then you know a block of some sort is present, but we will get to that later.

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ST SEGMENT

The ST segment is the short segment that connects the QRS complex to the T wave.

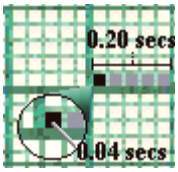
Normally, the ST segment is flat. Under conditions where the heart is damaged, such as myocardial ischemia (decreased oxygen supply to the heart), or trauma to the heart tissue, the ST segment may become either depressed OR elevated.



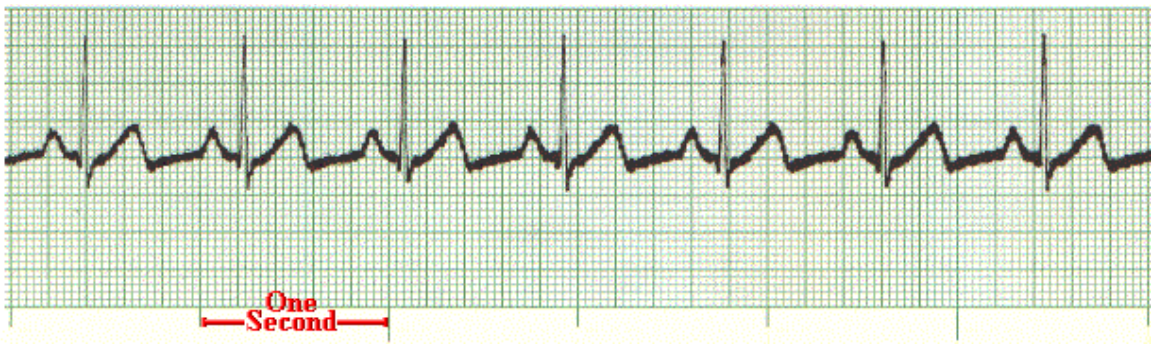
It may be helpful to measure how elevated and ST segment is when dealing with patients who have sustained a heart attack. In this program, however, we will only recognize whether the ST segment is elevated or depressed.



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STEP ONE: RATE



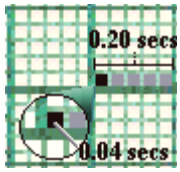
Calculating the rate is one of the easiest tasks in ECG interpretation. You'll notice at the bottom of the ECG that there are short green lines extending vertically from the grid-portion. These lines are one second markers. This means that the distance from one green line to the next is one second.

Looking at this ECG strip, you'll notice that it is divided into six blocks by these short green lines. Knowing that the distance between them is one second, it is not very hard to figure out that this ECG strip is six seconds long.

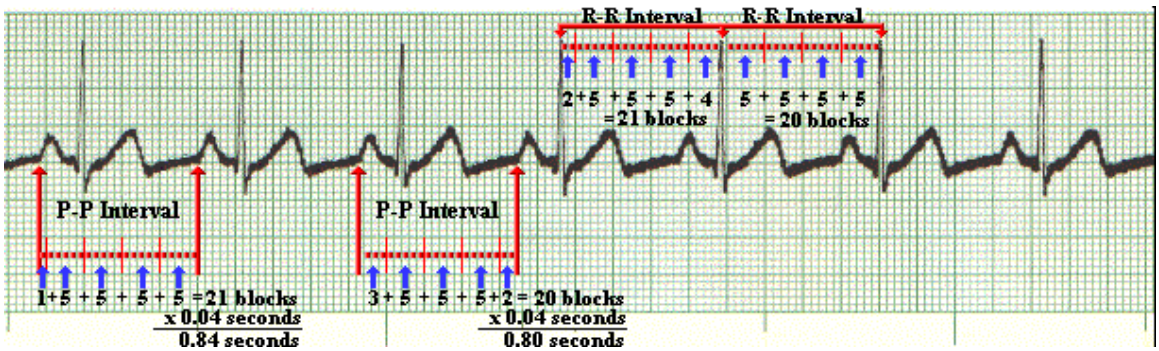
SINCE RATE IS COUNTED AS THE NUMBER OF BEATS PER MINUTE, YOU CAN SIMPLY COUNT THE NUMBER OF BEATS IN A SIX-SECOND STRIP AND MULTIPLY THIS NUMBER BY TEN.

In this six second strip, there are 7 beats. By multiplying this number by 10, we know that the heart rate is 70 beats per minute.

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STEP TWO: RHYTHM



As confusing as it may look, determining the rhythm of an ECG is really very simple. An ECG has an atrial rhythm and a ventricular rhythm. They can be either regular or irregular. All you have to do is determine whether or not the atrial and ventricular impulses occur regularly or not.

For example, to determine the atrial rhythm, you simply measure the distance from the beginning of one P wave to the beginning of the next P wave. By counting the number of small blocks between them, you accurately determine the distances and compare the two.

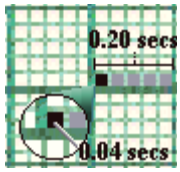
For an atrial rhythm to be considered regular, the distance between the two P waves needs to be within one or two small blocks of each other. As you see in the example above, the P-P interval (the distance between two P waves) is 21 blocks.

The next P-P interval measured is 20 blocks. Since these two values are within 2 blocks of each other, they are considered to be spaced regularly apart. If the P-P intervals are consistently spaced apart like this through the rest of the rhythm, it is considered to be regular. If the distance between them varies by more than 2 blocks, it is considered to be irregular.

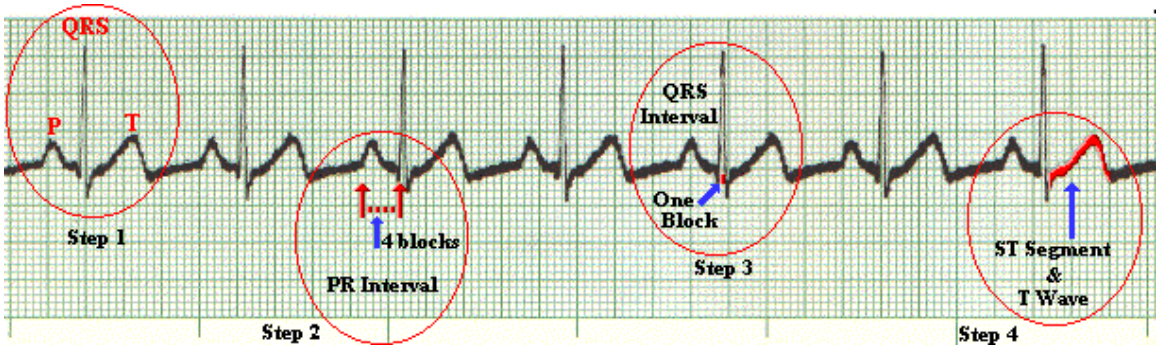
The ventricular rhythm is determined in the same fashion. By measuring the distance from R wave to R wave (the R-R interval) you can determine if the ventricles are creating a regular or irregular rhythm.

REMEMBER: Just because the atrial rhythm is regular , it does not mean that the ventricular rhythm will be regular and vice versa. Some abnormal rhythms, or arrhythmias, will have atrial and ventricular rhythms that are completely unrelated to each other.

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STEP THREE: CONDUCTION



There are actually four parts to Step Three, but it's easy to remember if you think of evaluating conduction as evaluating the waves, intervals and segments.

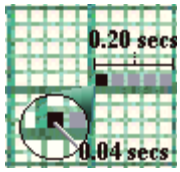
The first step in evaluating conduction is to look at the P,QRS, and T waves. Ask yourself if all three components are present. Also, check to see if all three are present in every complex in the strip.

Next, if P waves are present, check the PR interval. As we explore different arrhythmias, this will help you find blocked conductions.

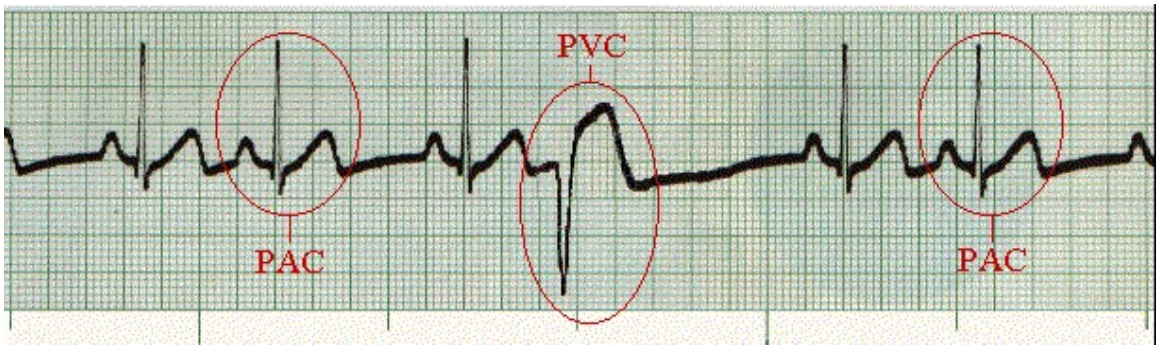
The third step is to examine the QRS interval, or the width of the QRS. Is it wider than 3 blocks? As you will see later, beats that originate in the ventricles have a QRS width greater than 3 blocks.

The fourth step is to examine the ST segment and T wave. Is the ST segment normal, or is depressed or elevated? Is the T wave upright or inverted?

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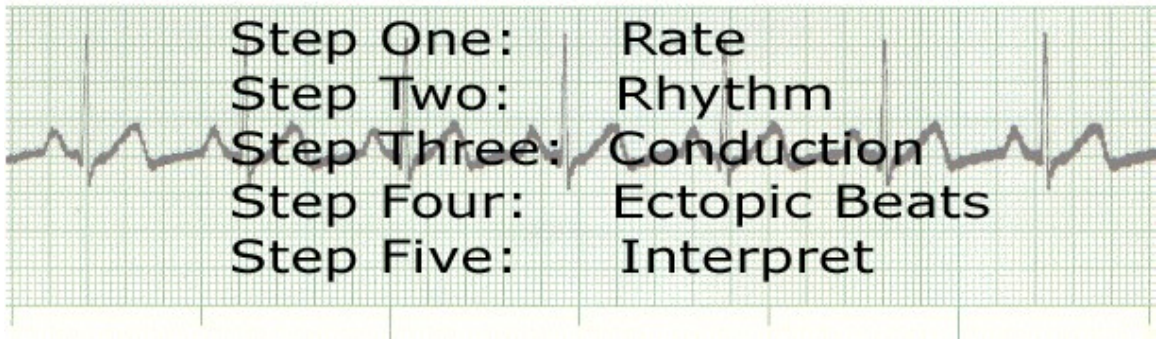
STEP FOUR: ECTOPIC BEATS



Ectopic beats are beats that originate from a focus other than the sinus node. Ectopic beats are usually shaped differently from the normal sinus beats and they usually occur prematurely. This means that they appear sooner than the normal beats and can be identified by the irregular rhythm they produce.

The two most common types of ectopic beats which we will examine are premature ventricular contractions (PVC's) and premature atrial contractions (PAC's).

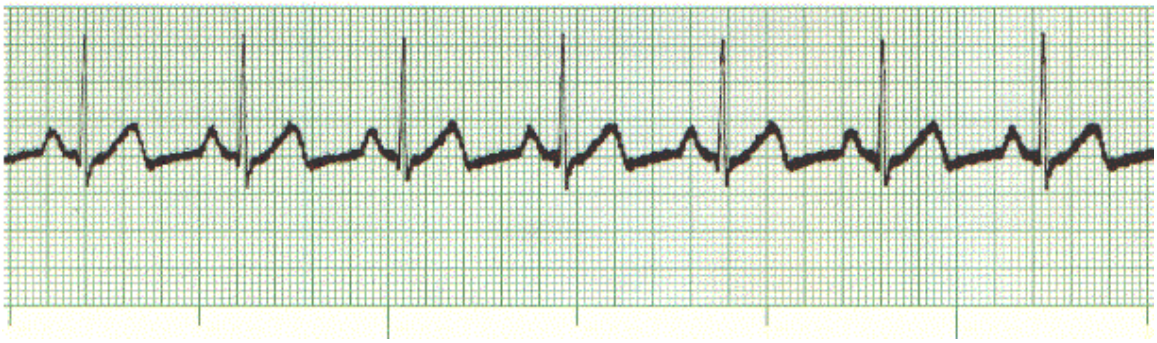
STEP FIVE: INTERPRET YOUR FINDINGS



The last step in rhythm identification is analyzing the information you have gathered and interpreting it. As we go through each of the different rhythms, we will take each step one-by-one and explain the interpretation process. Then you will be ready to tackle the practice rhythms on your own.

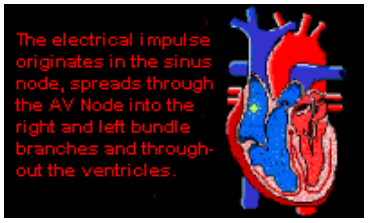
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NORMAL SINUS RHYTHM



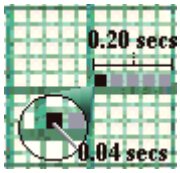
Step 1: Determine the rate. As you look at the strip, you'll notice that there are short, green, vertical lines extending from the bottom of the grid that divide the strip into six sections. As we indicated earlier, this means that this is a six-second strip. By counting the number of beats on the strip and multiplying by ten, you will have the rate. The rate of this strip is 70 beats per minute. Normal sinus rhythm is characterized by a rate of 60 to 100 beats per minute.

CONDUCTION NOTES



Step 2: Determine the rhythm. There are about 21 small blocks between each R wave and 21 small blocks between each P wave. This indicates that both the atrial and ventricular rhythm are regular. Normal sinus rhythm will always have both regular atrial and ventricular rhythms.

Step 3: Evaluate conduction. PQRST - Each beat has a P,QRS, and T wave and they are shaped the same throughout the rhythm strip which means the impulse begins in the sinus node and travels the normal electrical route. PR Interval - There are 4 small blocks (0.16 secs) between the beginning of the P wave and the beginning of the QRS. This means there is no block between the sinus node and the AV node. Normal sinus rhythm will always have a PR interval less than 5 small blocks (0.20 secs). QRS interval - The QRS interval is only 1 small block (0.04 secs). This reaffirms that the beat did not originate within the ventricles. ST segment & T wave - The ST segment is even with the baseline and the T wave is upright. Although both of these are normal findings, you can have alterations of the ST segment or T wave in normal sinus rhythm.

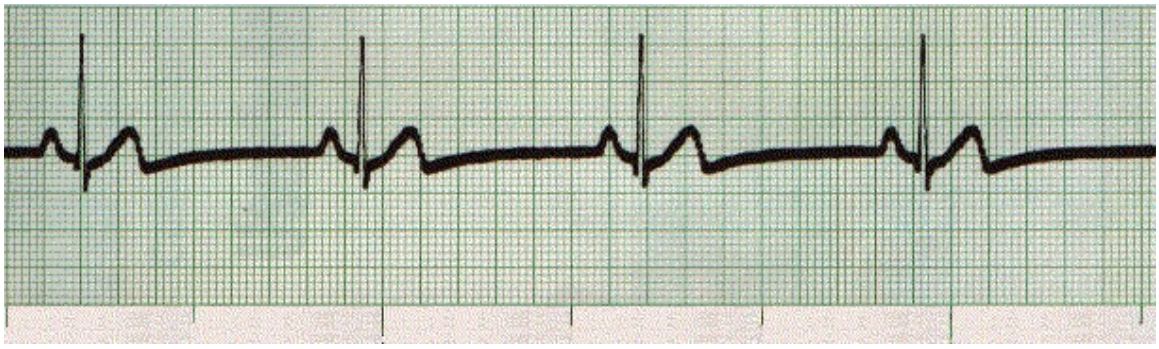


Step 4: Check for ectopic beats. Look carefully for abnormally-shaped complexes and complexes that appear too early or too late. All of these complexes are shaped the same and are regular in both atrial and ventricular rhythm, so there are no ectopic beats in this strip. Remember: You can sometimes find ectopic beats in this or any other rhythm strip.

Step 5: Interpret your findings. Of course, this is fairly easy since the name of the strip is at the top of your screen, but you need to get in the habit of this so you can identify strips later. By noting the presence of the normal P wave you know that rhythm originated in the sinus node, which means this is a sinus rhythm. The QRS complex, T wave and all associated intervals are normal which means there are no blocked conductions and there are no ectopic beats. Since the rate is between 60 and 100, you know that this normal sinus rhythm.

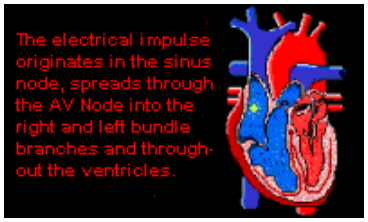
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SINUS BRADYCARDIA



Step 1: Determine the rate. By multiplying the number of beats in this six-second strip by ten, we get a rate of 40 beats per minute.

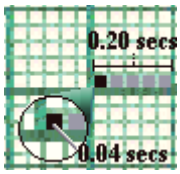
CONDUCTION NOTES



Step 2: Determine the rhythm. The intervals between each P wave are equal, so the atrial rhythm is regular. The intervals between the R waves are also equal so the ventricular rhythm is also regular.

Step 3: Evaluate conduction. PQRST - There are P, QRS, and T Waves in every beat, and they are consistently shaped throughout the strip. PR interval - The PR interval is 4 1/2 small blocks (0.18 secs) to 5 small blocks (0.20 secs). QRS interval - The QRS interval is less than 1 small block (0.04 sec) in width. ST Segment and T Wave - The ST segment is not elevated or depressed and the T wave is upright.

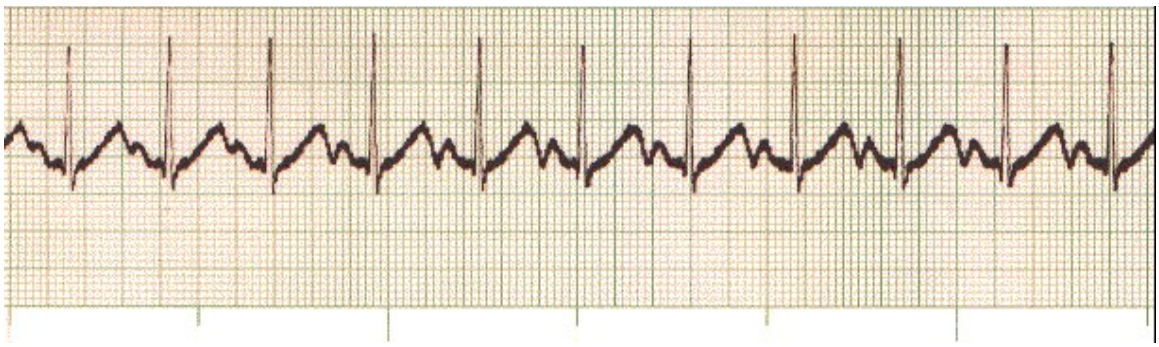
Step 4: Check for ectopic beats. There are no ectopic beats in this rhythm, but it is very common to see them in slow rhythms like this one.



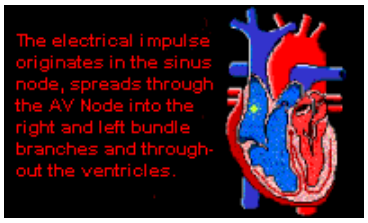
Step 5: Interpret your findings. The P wave in every complex indicates that this is a sinus rhythm. The atrial and ventricular rhythm is regular. The PR interval, ST segment, QRS interval, and T wave are all normal, and there are no ectopic beats, but the rate is 40 beats per minute, which is slow. A sinus rhythm that is less than 60 beats per minute is called BRADYCARDIA. So this rhythm is called sinus bradycardia.

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SINUS TACHYCARDIA



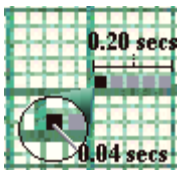
CONDUCTION NOTES



Step 1: Determine the rate. When you see that there are incomplete beats on either end of the six-second strip, it is easiest to calculate the rate by simply counting the number QRS complexes you see. In this strip, there are 11 QRS complexes, so the rate is 110 beats per minute.

Step 2: Determine the rhythm. The P waves are about 14 small blocks (0.56 secs) from each other, so the atrial rhythm is regular. The QRS complexes are also about 14 small blocks (0.56 secs) from each other, so the ventricular rhythm is regular as well.

Step 3: Evaluate conduction. PQRST - There are P,QRS and T waves for every beat and the P waves are all similar in size and shape, so this rhythm originates in the sinus node. PR Interval - The PR interval for this rhythm is about 4 small blocks (0.16 secs) which is normal. QRS Interval - The width of the QRS complex is one small block (0.04 sec) which is normal. ST Segment and T Wave - The ST segment is not elevated or depressed, and the T wave is upright.

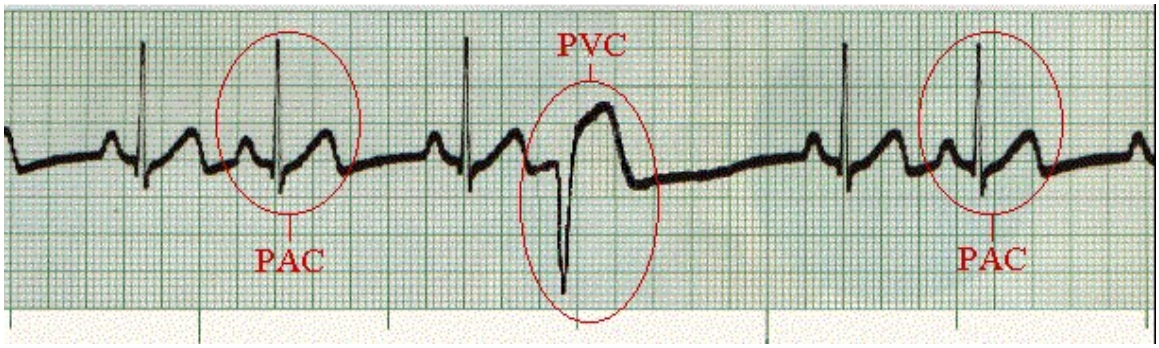


Step 4: Check for ectopic beats. All of the beats are similiar in size, shape and configuration and they are all spaced evenly apart. There are no ectopic beats in this rhythm.

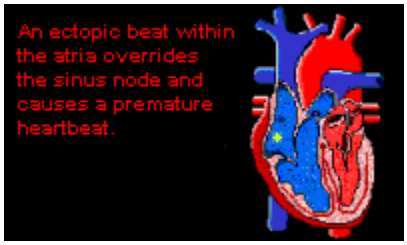
Step 5: Interpret your findings. This is a sinus rhythm that has a rate of 110 beats per minute. All of the other conduction measurements are normal. A sinus rate faster than 100 beats per minute is called a TACHYCARDIA, and so, this rhythm is called sinus tachycardia.

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PREMATURE ATRIAL CONTRACTION

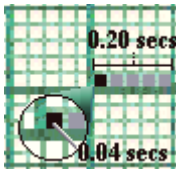


CONDUCTION NOTES



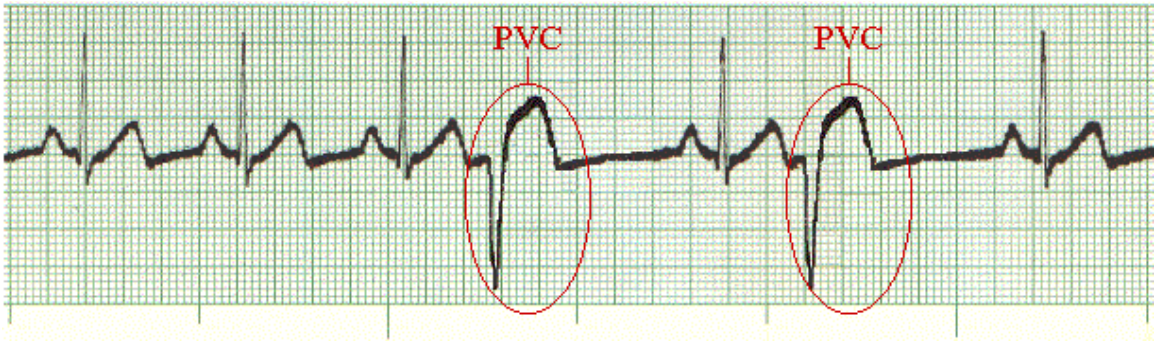
Premature atrial contractions (PAC's) are abnormal beats that, when they occur, are usually in the setting of normal sinus rhythm. These beats originate from an irritable focus in the atria, often very close to the sinus node. At First, PAC 's may be hard to detect because they look so much like the normal sinus beats around them. If you look closely, though, you'll notice that the beat appears earlier than the others and makes both the atrial and ventricular rhythm irregular. Also, since the impulse does not originate from the sinus node like the beats around it, the P wave sometimes has a slightly different shape.

The rhythm above also demonstrates an example of a PVC, which we will study next. Be sure to look back at this page to review the difference in appearance of PACs and PVCs.



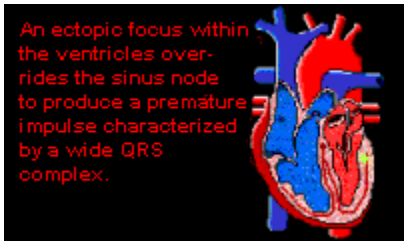
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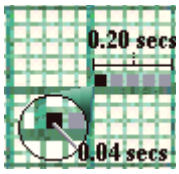
PREMATURE VENTRICULAR CONTRACTION



Premature ventricular contractions (PVC's) are beats that originate in the ventricles and cause an irregularity due to their early apppearance. Beats that originate in the ventricles are easy to spot because of their characteristic wide and bizarre shape.

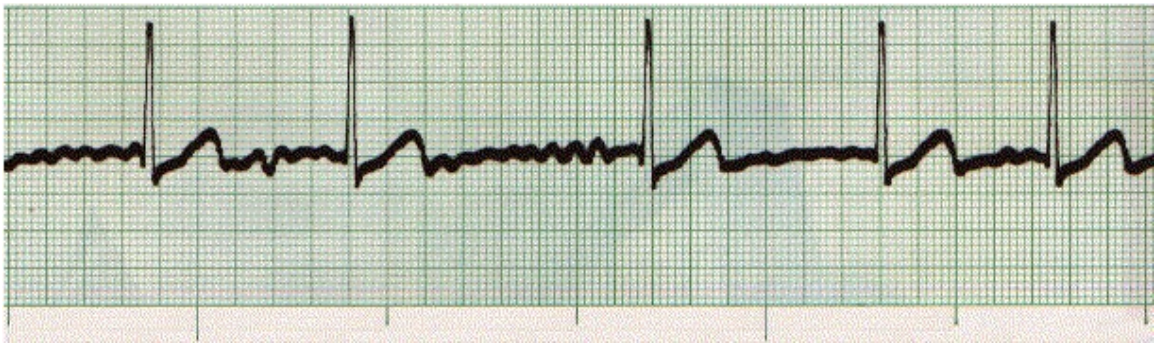
CONDUCTION NOTES



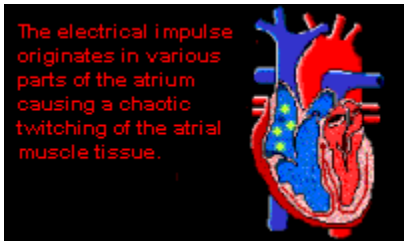


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ATRIAL FIBRILLATION



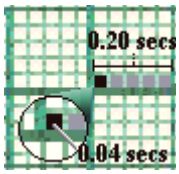
CONDUCTION NOTES



Step 1: Determine the rate. Generally, when someone refers to the heart rate, they are referring to the ventricular rate, because it more closely indicates how many times a minute the heart is pumping blood into the aorta. There are 5 QRS complexes in this six-second strip, so the ventricular rate is 50 beats per minute. There are dozens of P waves, or atrial contractions, however, and some of them are impossible to even measure. The atrial rate in this rhythm can be as high as 400 per minute, but many times, the amplitude (height) of the waves is very low and they appear too blended into the baseline to count accurately. Sometimes, the ventricles will respond to the rapid electrical activity of the atria and the ventricular rate will be as high as 150 beats per minute, or even higher. People with this rhythm usually take medications to control the rate of ventricular contractions.

Step 2: Determine the rhythm. One of the most noticeable characteristics about this rhythm is that the P-P intervals and the R-R intervals are completely unpredictable. Both the atrial and ventricular rhythms are said to be irregular.

Step 3: Evaluate conduction. PQRST - Obviously, atrial fibrillation is a conduction problem within the atria. The numerous, oddly-shaped P waves indicate that the impulses are originating all over the atria very rapidly. Most of these impulses are not strong enough to stimulate any tissue except for a small amount of cardiac muscle around it. This causes the atria to fibrillate, which means to quiver. The QRS complexes, although normal in size and shape, appear irregularly as impulses are conducted through, or even by the AV Node in an unpredictable fashion. The T waves appear normally after the QRS complexes.



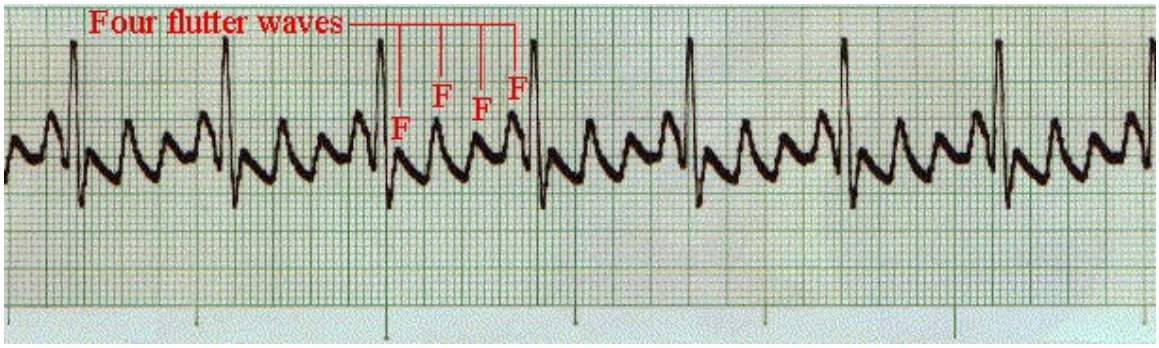
Step 4: Check for ectopic beats. Despite the irregularity of the P waves and the unpredictable ventricular rate, there are no ectopic beats in this rhythm.

Step 5: Interpret your findings. Atrial fibrillation is easy to identify because of it's irregular and unpredictable atrial and ventricular rate. It has normal QRS complexes and T waves, but the atrial impulses originate in many different places rather than the sinus node which causes the P waves to have different sizes and shapes.

DIAGNOSTIC TOOL: Atrial fibrillation has an irregular ventricular rhythm and P waves that unpredictable both in rhythm and shape.

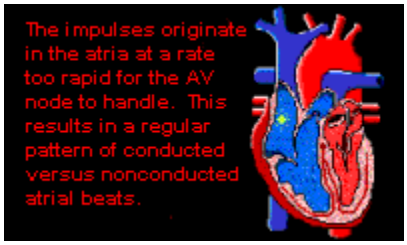
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ATRIAL FLUTTER



Step 1: Determine the rate. This strip may seem a little different from the others we have looked at, and it is. Notice how there are 4 sawtooth-shaped waves between each QRS? Those are the indication of atrial activity, but they are not called P waves. Instead, they are called flutter waves. These sawtooth waves are an identifying characteristic of atrial flutter. There are 26 of them in this strip, which means the atrial rate is 260 beats per minute. There are 7 QRS complexes, which means that the ventricular rate is 70 beats per minute

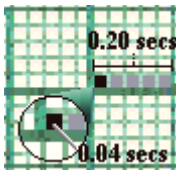
CONDUCTION NOTES



Step 2: Determine the rhythm. Another identifying characteristic of atrial flutter is that the atrial and ventricular rhythms are both regular. Notice that the flutter waves are spaced evenly apart and, although they are not identical, the waves appear similar from beat to beat. Now think about how atrial fibrillation looked and how chaotic and unpredictable the P waves looked.

Step 3: Evaluate conduction. PQRST - There are no P waves in the rhythm, instead, there are four flutter waves for each QRS. Remember, though, that the flutter waves still represent atrial depolarization. There is a normal QRS complex for each beat, but there are no identifiable T waves, only flutter waves. PR Interval - Since there are no P waves, the PR interval is not measurable. QRS interval - The QRS interval is about 1 1/2 small blocks (0.06 secs) which is normal. ST Segment and T Wave - Without identifiable T waves, neither of these components can be measured.

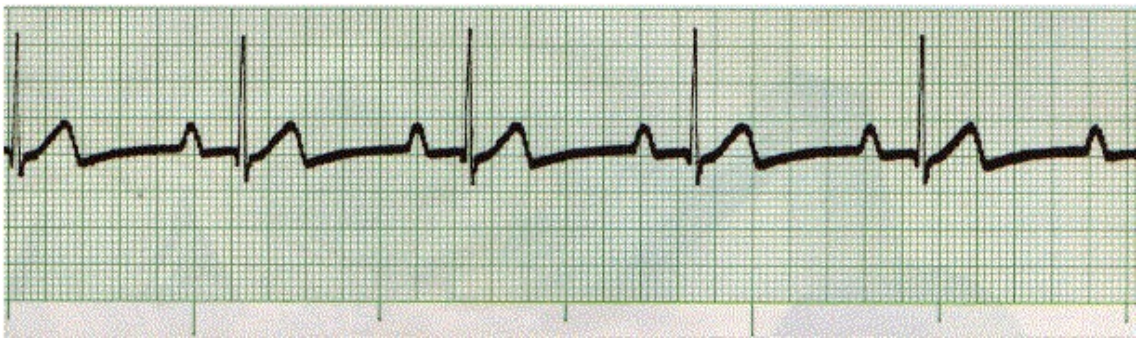
Step 4: Check for ectopic beats. There are no ectopic beats in this rhythm.



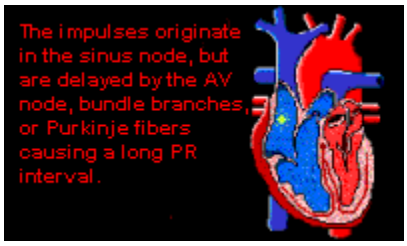
Step 5: Interpret your findings. This rhythm originates in the atria, but because of a malfunction that is probably occurring in the atrial conduction system, the atria are depolarizing very rapidly. The AV Node cannot keep up with the atrial rate of 260, so it is conducting the impulses that it is able, which in this circumstance is every fourth impulse. Sometimes, in atrial flutter, you will see a QRS for every 2,3,4 or even five flutter waves. Usually, the ventricular rhythm is regular, but on some occasions you will see variable conduction, which means that the AV node will alternate the number of flutter waves it takes to produce a QRS with each beat. The rhythm is still identifiable by the characteristic flutter waves.

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FIRST DEGREE BLOCK



CONDUCTION NOTES

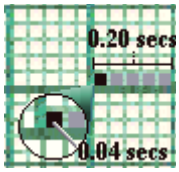


Step 1: Determine the rate. The atrial and ventricular rates are regular. There are 5 QRS complexes on this strip, so the rate is 50 beats per minute.

Step 2: Determine the rhythm. The P waves are spaced evenly apart, and so are the QRS complexes, therefore, the atrial and ventricular rhythms are regular.

Step 3: Evaluate conduction. PQRST - There are P, QRS, and T Waves for every complex. PR Interval - The PR interval is about 8 small blocks (0.32 secs) long. Remember that the normal PR interval is not longer than 5 small blocks (0.20 secs). This prolonged PR interval is the defining feature of first degree block. QRS Interval - The QRS interval is one small block (0.04 secs) wide and is normal. ST Segment and T Wave - The ST segment is not elevated or depressed and the T wave is upright and normal.

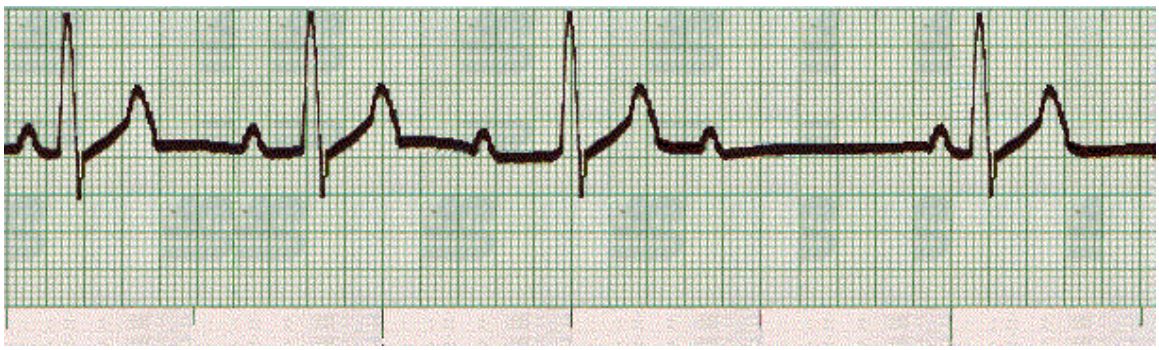
Step 4: Check for ectopic beats. There are no ectopic beats in this rhythm.



Step 5: Interpret your findings. This rhythm originates in the atria, but because of a delay within the AV node, the PR interval is longer than normal. A sinus rhythm with no abnormalities other than a prolonged PR interval is identified as first degree block.

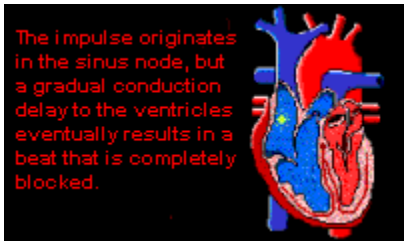
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SECOND DEGREE BLOCK TYPE I



Step 1: Determine the rate. There are 5 P waves, so the atrial rate is 50 beats per minute and there are 4 QRS complexes, so the ventricular rate is 40 beats per minute.

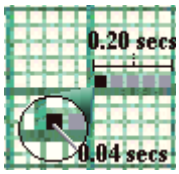
CONDUCTION NOTES



Step 2: Determine the rhythm. The P waves are evenly spaced at about 31 small blocks apart (1.24 secs) which indicates that the atrial rhythm is regular. The R-R interval, or the distance from one QRS complex to the next increases with each beat in a cycle, so the ventricular rhythm is irregular.

Step 3: Evaluate conduction. PQRST - There are P, QRS, and T Waves in every complex except for one, where there is a P wave, but no QRS complex or T wave after it. PR interval - The PR interval is the key to understanding second degree block type I.

The PR interval lengthens with each beat until finally a complete block forms and no QRS complex follows the P wave, then the cycle of lengthening PR intervals begins again. For example, the PR interval in the first beat is 5 1/2 small blocks (0.22 secs). This is a prolonged PR interval already. The second beat, though, has a PR interval of 8 small blocks (0.32 secs) which is even longer! Ther third beat has a PR interval of 12 small blocks (0.48 secs) and the next P Wave does not even produce a QRS complex. After the beat is dropped, the cycle starts all over again with a PR interval of 5 1/2 small blocks (0.22 secs) and repeats itself. QRS Interval - The width of the QRS is 2 small blocks (0.08 secs) which is normal. ST Segment and T Wave - The ST segment is neither elevated or depressed and the T wave is upright (normal).

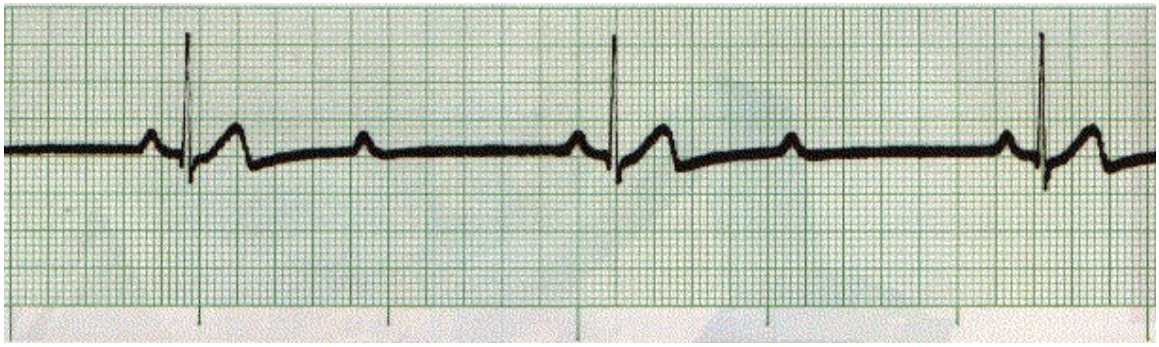


Step 4: Check for ectopic beats. There are no ectopic beats in this rhythm strip.

Step 5: Interpret your findings. The lengthening of the PR interval over two or more beats and eventual complete block of a QRS occurring in a repeated cycle is the primary diagnostic indicator of second degree block type 1.

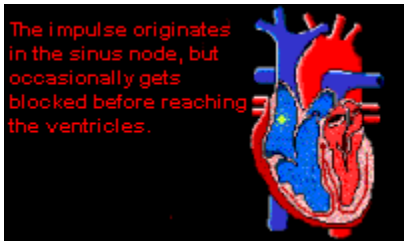
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SECOND DEGREE BLOCK TYPE II



Step 1: Determine the rate. The 5 P waves and 3 QRS complexes in this strip tell you that the atrial rate is 50 and the ventricular rate is 30.

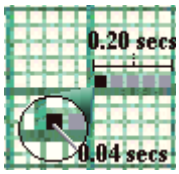
CONDUCTION NOTES



Step 2: Determine the rhythm. The P waves are all evenly spaced at about 28 small blocks (1.12 secs) apart, so the atrial rhythm is regular. The QRS complexes are evenly spaced at about 57 small blocks (2.28 secs) apart, so the ventricular rhythm is also regular.

Step 3: Evaluate conduction. PQRST - Every other P wave has a normal QRS complex and T wave following it, but the other P waves are alone. PR Interval - Only the three beats with P,QRS, and T Waves have a measurable PR interval. The interval on these beats is about 5 small blocks (0.20 secs), which is normal. QRS Interval - The QRS interval is about 1 small block (0.04 secs) wide, which is normal. ST Segment and T Wave - The ST segment is not elevated or depressed in this rhythm, so it is normal. The T wave is upright and also normal.

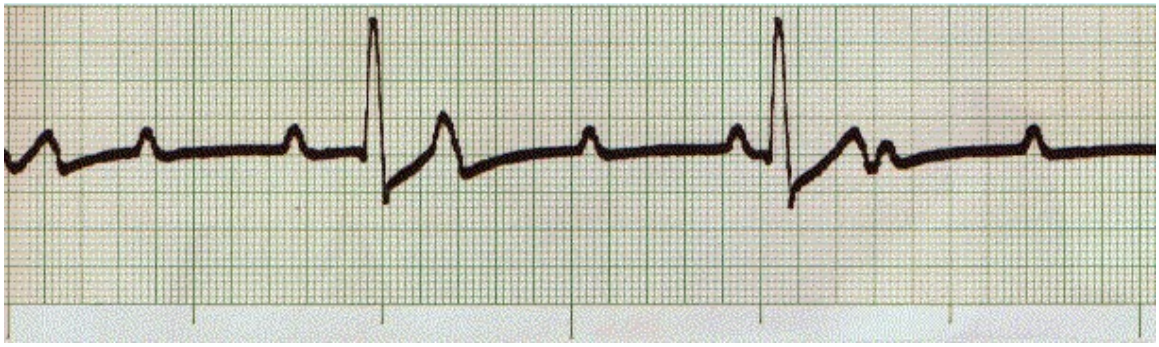
Step 4: Check for ectopic beats. There are no ectopic beats in this rhythm.



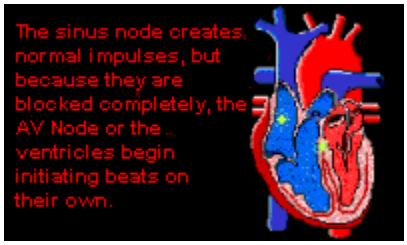
Step 5: Interpret your findings. This rhythm is characterized by a regular atrial rhythm, constant and usually normal PR interval. The abnormality in this rhythm that is easiest to identify is the occasional nonconduction of a P wave. This can happen in a regular rhythm as it appears in this strip, or it can occur irregularly. The chief difference between type 1 second degree block and type 2 is that the PR interval stays constant in Type 2, whereas it progressively lengthens in type 1.

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THIRD DEGREE BLOCK



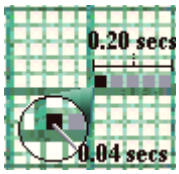
CONDUCTION NOTES



Step 1: Determine the rate. There are 5 P waves in this six-second strip, so it is easy to determine that the atrial rate in 50 beats per minute. There are 4 QRS complexes which indicate that the ventricular rate is 40 beats per minute. Remember that the ventricular rate represents the number of ventricular contraction per minute, or heart rate.

Step 2: Determine the rhythm. There are consistently about 31 small blocks (1.24 secs) between each P wave, so the atrial rhythm is regular. There are about 34 small blocks (1.36 secs) between each QRS complex, so the ventricular rhythm is also regular.

Step 3: Evaluate conduction. PQRST - P, QRS, and T Waves are all present in this rhythm. If you look closely, though, you will notice that the P waves have no relationship at all with the QRS complex and T wavei. This indicates that the atria are beating independently of the ventricles.Although, the sinus node is firing properly, the impulse is completely blocked before it can stimulate the ventricles. This is why Third Degree Heart Block is sometimes referred to as Complete Heart Block. PR Interval - Because there is no conduction relationship with the QRS, the PR interval varies with each beat. QRS Interval - The QRS interval is 2 1/2 small blocks (0.10 secs) wide. Because the atrial impulse is completely blocked, the AV Node began firing spontaneously on it's own in order to keep the heart beating. Sometimes, the ventricles themselves will begin firing spontaneously instead of the AV node. The way you determine which is firing is by examining the QRS interval. If the QRS interval is less than or equal to 3 small blocks (0.12 secs) then the impulse is originating in the AV node. If the QRS interval is greater than 3 small blocks (0.12 secs), then the impulse originated within the ventricles. ST Segment and T Wave - The ST segment is not depressed or elevated and the T wave is upright (normal).

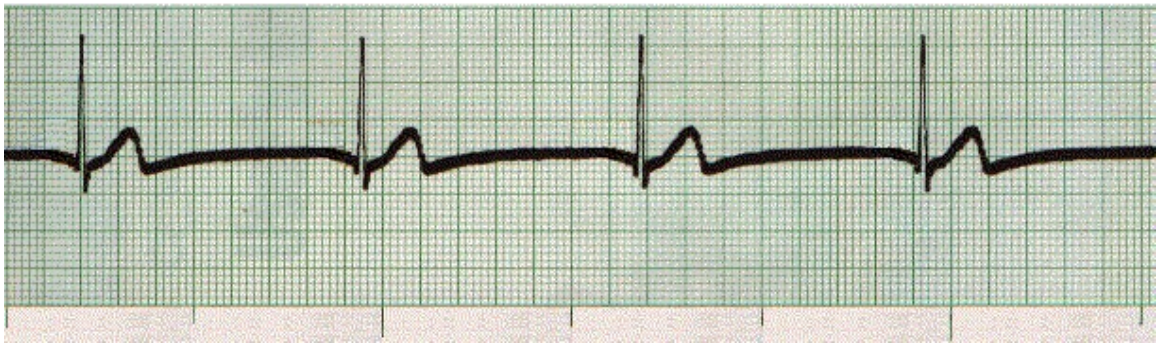


Step 4: Check for ectopic beats. Actually, because none of the atrial impulses are being conducted, all of the QRS complexes represent ectopic junctional beats. That is to say, they are all beats that originated in the AV Node as a safety mechanism to keep the heart beating.

Step 5: Interpret your findings. A regular P-P Interval, regular R-R interval, and lack of association between the P waves and QRS complexes are the identifying characteristics that tell you this is third degree block.

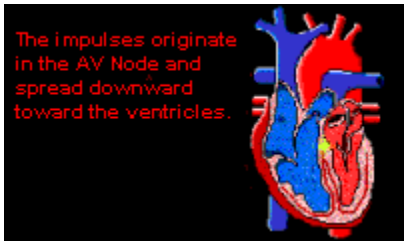
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JUNCTIONAL (OR NODAL) RHYTHM



Step 1: Determine the rate. In this rhythm, no P waves are noticeable, so we won't worry about the atrial rate for now. There are 4 QRS complexes, though, so the ventricular rate is 40.

CONDUCTION NOTES

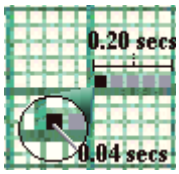


Step 2: Determine the rhythm. The QRS complexes are evenly spaced at about 37 small blocks apart (1.48 secs), so the ventricular rhythm is regular. Although you can't see any P waves in this strip, the atrial rhythm is also regular, but we'll discuss that in the interpretation section.

Step 3: Evaluate conduction. PQRST - There is a normal QRS complex and T wave for every beat, but there are no obvious P waves. PR Interval - Unable to determine. QRS Interval - The QRS interval is 1 small block (0.04 secs) wide, which is normal. ST Segment and T Wave - There is no ST segment depression or elevation, which is normal. The T wave is upright and normal.

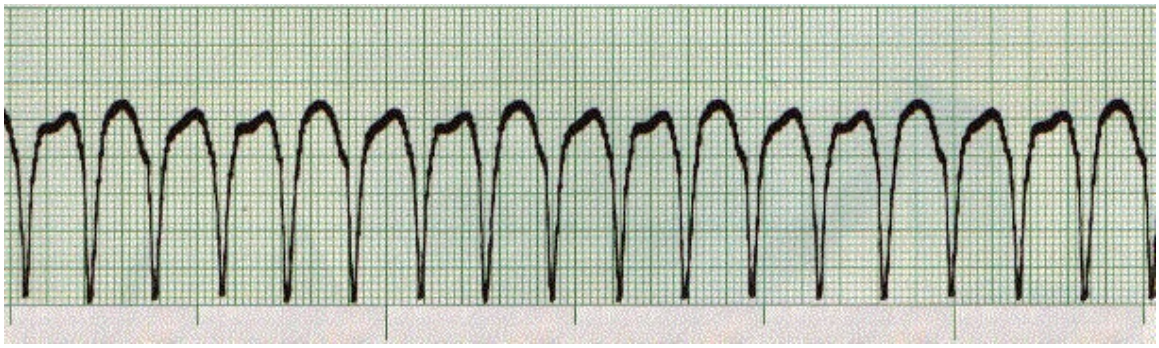
Step 4: Check for ectopic beats. There are no PAC's or PVC's in this rhythm, but every beat originates in the AV node and could technically be considered an ectopic rhythm.

Step 5: Interpret your findings. Junctional rhythms originate from the AV Node and is sometimes called a nodal rhythm. The rate is normally 40-60 beats per minute and the ventricular rhythm is regular. As the AV node fires, the impulse travels down towards the ventricles to produce the QRS, and upward toward the atria. Because the atria are stimulated in this abnormal manner, the P wave may appear upside down, after the QRS or even not at all, as in this rhythm. When it does appear, though, the atrial rhythm is regular.



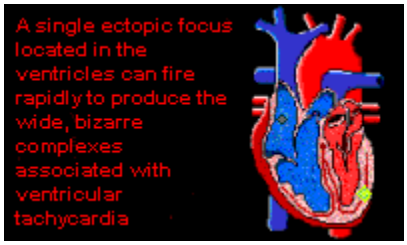
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VENTRICULAR TACHYCARDIA



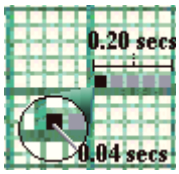
Step 1: Determine the rate. No P waves are visible, so the atrial rate cannot be determined. There are 18 QRS complexes present in this strip, so the ventricular rate is 180 beats per minute.

CONDUCTION NOTES



Step 2: Determine the rhythm. Without visible waves, it is impossible to determine atrial rhythm, but by measuring the R-R Interval it can be determined that the QRS complexes are evenly spaced at about 8 small blocks (0.32 secs) apart. Thus, the ventricular rhythm is regular.

Step 3: Evaluate conduction. PQRST - There are no discernible P waves in this rhythm, and the QRS complex and T wave appear to be merged together. PR interval - The PR interval cannot be measured on this strip. QRS interval - The QRS interval is about 4 small blocks (0.16 secs) apart which is wider than normal. ST Segment and T Wave - The ST segment and T wave both are difficult to assess in this rhythm because the T wave appears merged with the QRS complex.

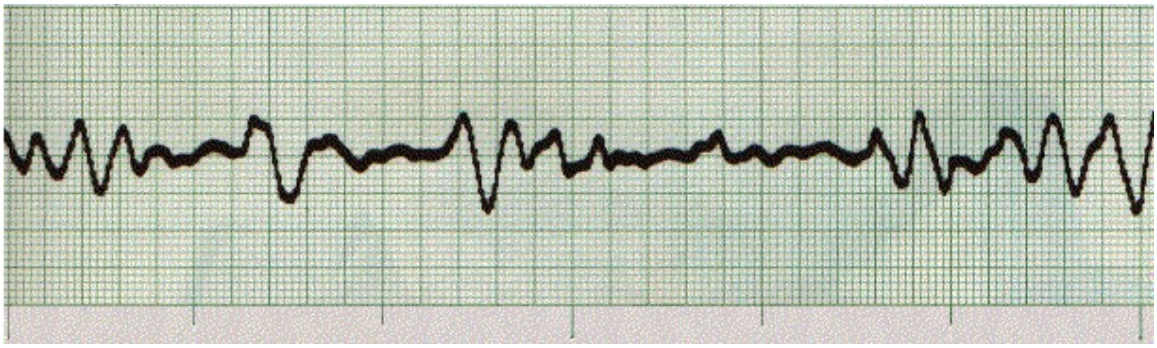


Step 4: Check for ectopic beats. Like the junctional rhythm, this rhythm is composed entirely of beats originating outside of the sinus node. and could technically be considered an ectopic rhythm.

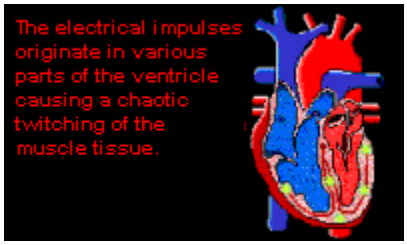
Step 5: Interpret your findings. Ventricular tachycardia is another easily recognized rhythm characterized by the rapid succession of wide ventricular beats.

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VENTRICULAR FIBRILLATION



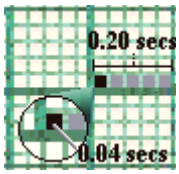
CONDUCTION NOTES



Step 1: Step 1: Determine the rate. Because there are no discernible P waves or QRS complexes, the atrial and ventricular rates cannot be determined.

Step 2: Determine the rhythm. Without discernible P waves or QRS complexes, the atrial and ventricular rhythms cannot be determined.

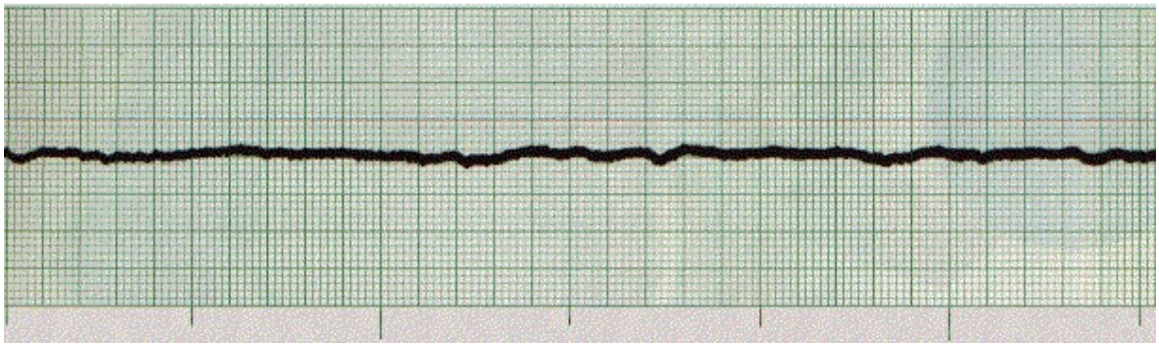
Step 3: Evaluate conduction. PQRST - There are no definable P, QRS or T Waves in this rhythm. PR interval - The PR interval cannot be determined. QRS interval - The QRS interval cannot be determined. ST segment and T wave - These cannot be evaluated.



Step 5: Interpret your findings. The conduction in ventricular fibrillation is characterized by the appearance of chaotic fibrillatory waves instead of the normal P,QRS and T Waves. This is caused by numerous electrical impulses forming within the ventricles which stimulate only a small area of tissue around it. This causes the ventricular muscle to quiver rather than contract. The loss of ventricular contraction means that the heart is not pumping any blood.

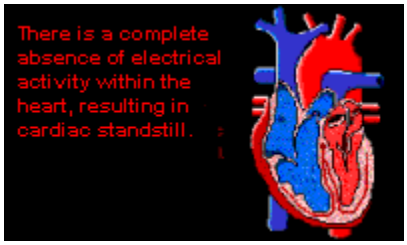
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ASYSTOLE

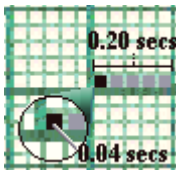


Asystole is the easiest rhythm to identify...because there isn't one. In asystole, there is a complete lack of any electrical activity within the heart, so the ECG simply shows a flat line. Of course, there is no muscular activity within the heart either.

CONDUCTION NOTES



IMPORTANT: Always check to make sure that the ECG leads are on and placed correctly before you make up your mind that someone's heart has stopped!



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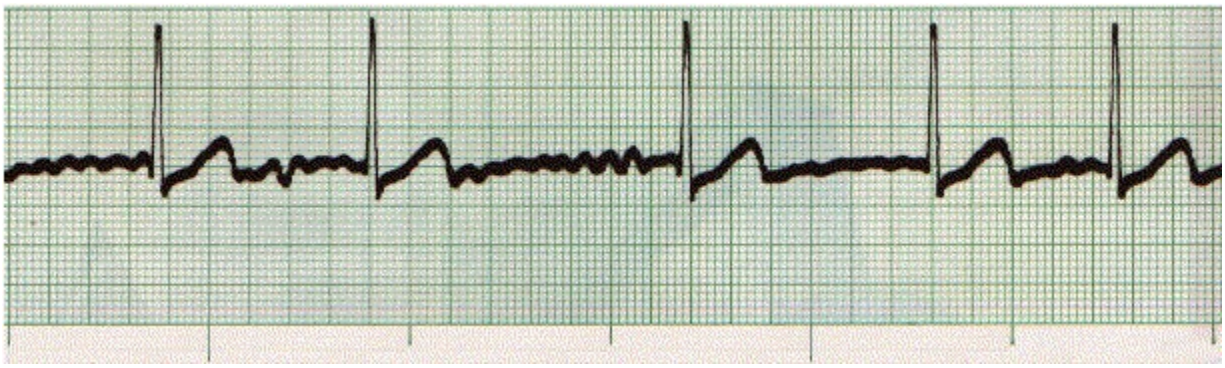
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Answer Sheet

1	A	B	C	D
2	A	B	C	D
3	A	B	C	D
4	A	B	C	D
5	A	B	C	D
6	A	B	C	D
7	A	B	C	D
8	A	B	C	D
9	A	B	C	D
10	A	B	C	D
11	A	B	C	D
12	A	B	C	D
13	A	B	C	D
14	A	B	C	D
15	A	B	C	D

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PROBLEM 1



T.L. is a eighty-two year old female who has been admitted to the telemetry unit with an exacerbation of congestive heart failure. She is receiving oxygen by nasal cannula at 3 L/min and her blood pressure is 114/68.

She received IV furosemide on arrival five hours ago. She states that she is still slightly short of breath, but you notice that she appears much calmer now. T.L. states that she has a prescription for digoxin at home, but hasn't taken it in over a week. How would you interpret this rhythm?

A. Normal sinus rhythm with PVCs

B. Atrial fibrillation

C. Atrial flutter

D. Junctional rhythm

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PROBLEM 2



G.K. is a fifty-five year-old female who has been received in the post-anesthesia care unit following a cholescytectomy. She appears lethargic and confused, but you are not sure if it may be due to the effects of anesthesia or not. Her blood pressure is 82/58 and her pulse is thready. On physical exam, you notice that her skin is pale, cool, and clammy. How would you interpret this rhythm?

A. Sinus tachycardia

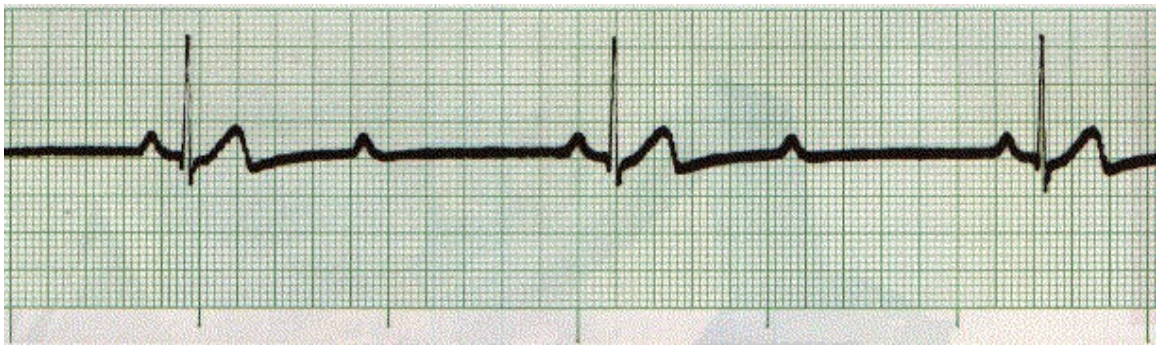
B. Ventricular fibrillation

C. First degree AV block

D. Normal sinus rhythm

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PROBLEM 3



W.S. is a 49 year-old male who has been admitted to the cardiac care unit following an outpatient thallium stress test. His physician suspects that he has advanced coronary artery disease and has scheduled him for a heart catheterization. W.S. denies any complaints right now, but states that he gets short of breath quickly on exertion. How would you interpret his cardiac rhythm?

A. Junctional rhythm

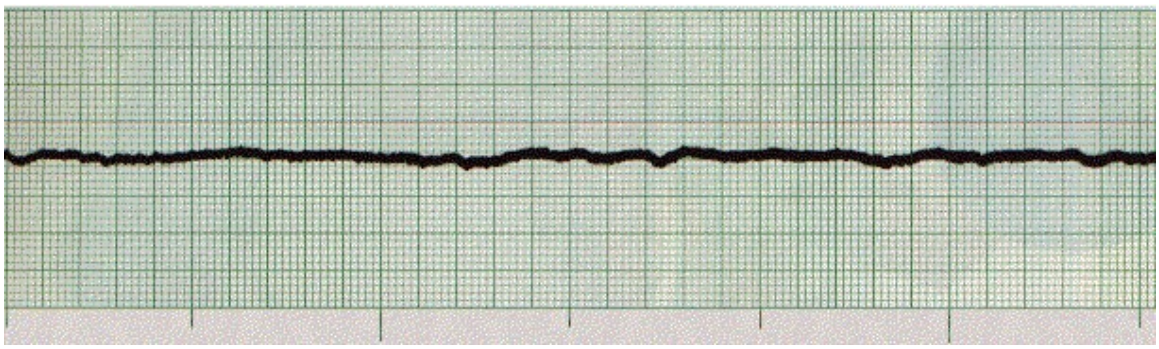
B. Atrial flutter

C. First degree block

D. Second degree AV block type II

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PROBLEM 4



J.M. is a seventy-six year old male who has arrived to the emergency department in cardiac arrest. After transferring him to the stretcher, the code team leader ordered compressions and bag-valve-mask respirations be stopped so that she could check for a pulse, rhythm, and spontaneous respirations. There is no pulse or spontaneous respirations. You have checked to ensure that the ECG leads are secured properly. How would you interpret this rhythm?

A. Sinus bradycardia

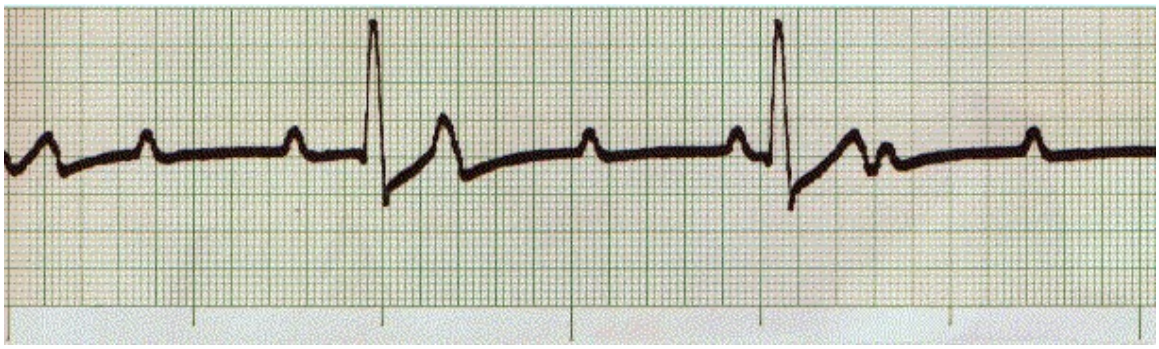
B. Asystole

C. Normal sinus rhythm with PACs

D. Ventricular tachycardia

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PROBLEM 5



T.J. is a 78 year-old female who has been admitted to the telemetry unit for observation after she passed out at home. She denies any pain or shortness of breath right now, but states that she feels faint when she stands up. You place on the cardiac monitor. How would you interpret this rhythm?

A. Third-degree AV block

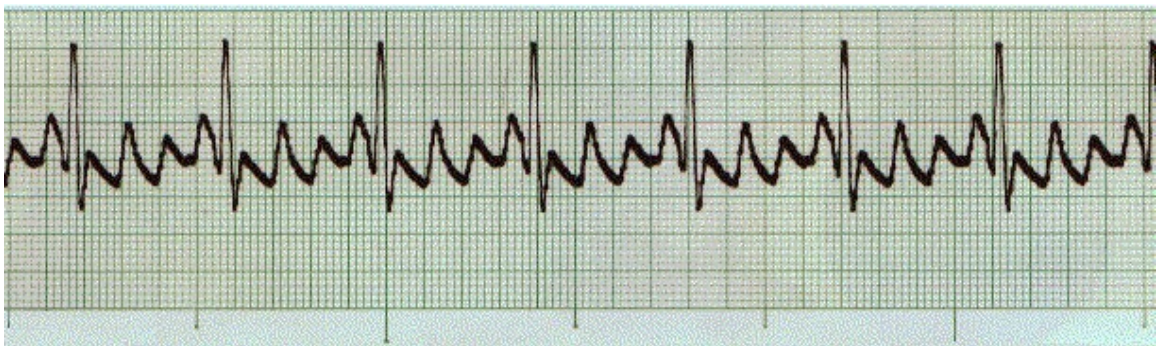
B. Normal sinus rhythm with PVCs

C. Second-degree AV block type I

D. Junctional rhythm

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PROBLEM 6



N.B. is a 32 year-old male who has arrived in the emergency department complaining of moderate dyspnea and sharp pleuritic chest pain of sudden onset. The paramedics who brought him state he coughed up a small amount of blood en route. You notice he is wearing a knee immobilizer. When questioned about it, he states that he had knee surgery about a month ago. His B/P is 144/68. How would you interpret this rhythm?

A. Sinus tachycardia

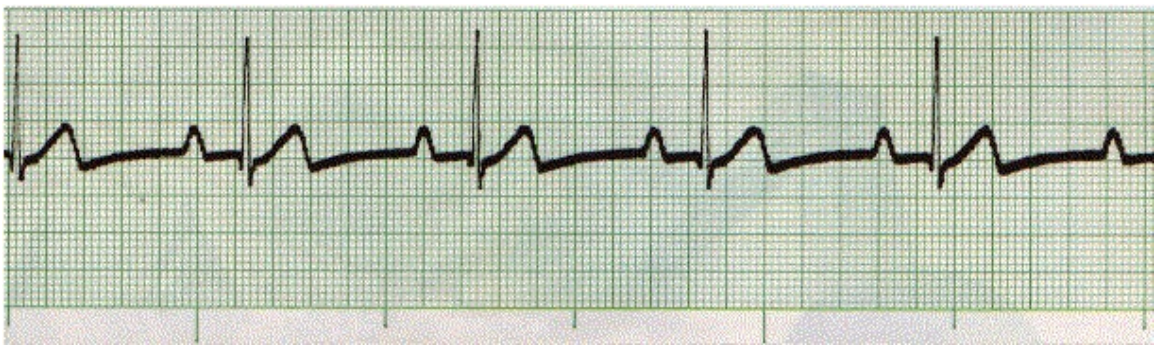
B. Ventricular fibrillation

C. Third degree block

D. Atrial flutter

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PROBLEM 7



G.T. is a 52 year-old female with a history of congestive heart failure who has been admitted for complaints of nausea, vomiting, diarrhea, and blurred vision. It was determined that her digoxin level was elevated. She has been admitted to the telemetry unit for observation and treatment. You hook her to the cardiac monitor. How would you interpret this rhythm?

A. First-degree AV block

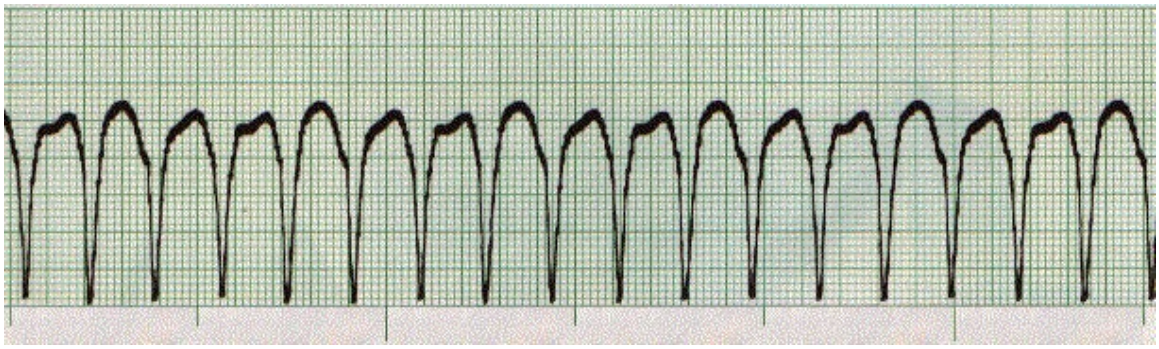
B. Second-degree AV block type I

C. Sinus bradycardia

D. Normal sinus rhythm

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PROBLEM 8



R.T. is a fifty-four year-old male who has arrived in the emergency department by ambulance complaining of substernal chest pain and severe shortness of breath. He is obviously anxious and you notice that his mouth and oral mucosa appear cyanotic. He has a rapid, thready pulse and his blood pressure is 94/40. How would you interpret this rhythm?

A. Ventricular fibrillation

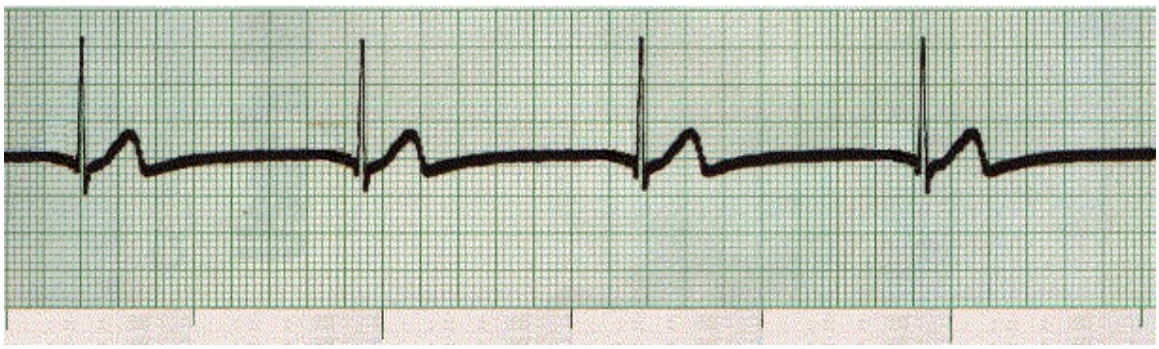
B. Sinus tachycardia

C. Ventricular tachycardia

D. Atrial fibrillation

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PROBLEM 9



N.B. is a 64 year-old female who has returned to the cardiac care unit following cardiac valve replacement surgery. She is still lethargic from the anesthesia, but her vital signs are stable. You notice that her current heart rhythm is different from the sinus tachycardia she was in prior to surgery. How would you diagnose this rhythm?

A. Sinus tachycardia

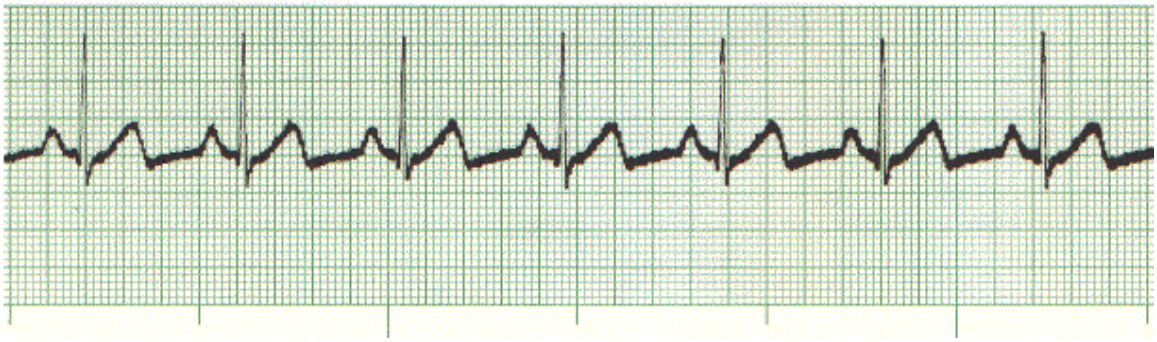
B. Junctional rhythm

C. Sinus bradycardia

D. First-degree AV block

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PROBLEM 10



H.G. is a 45 year-old female who has arrived in the emergency department complaining of a dull, substernal chest pain that started about 15 minutes after her last meal. She denies any shortness of breath or other symptoms. Her pulse is strong and regular and her blood pressure is 132/88. Initial cardiac enzymes are within normal limits. How would interpret this rhythm?

A. Normal sinus rhythm

B. Second-degree AV block type II

C. Sinus bradycardia

D. Junctional rhythm

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PROBLEM 11



T.H. is a sixty-two year old male with a history of two CABG operations and cardiomyopathy. He collapsed at home while watching television. When the paramedics arrived and placed him on the ECG monitor, this was the rhythm they noticed. The patient had no pulse or respirations and appeared cyanotic. How would you interpret this rhythm?

A. Atrial flutter

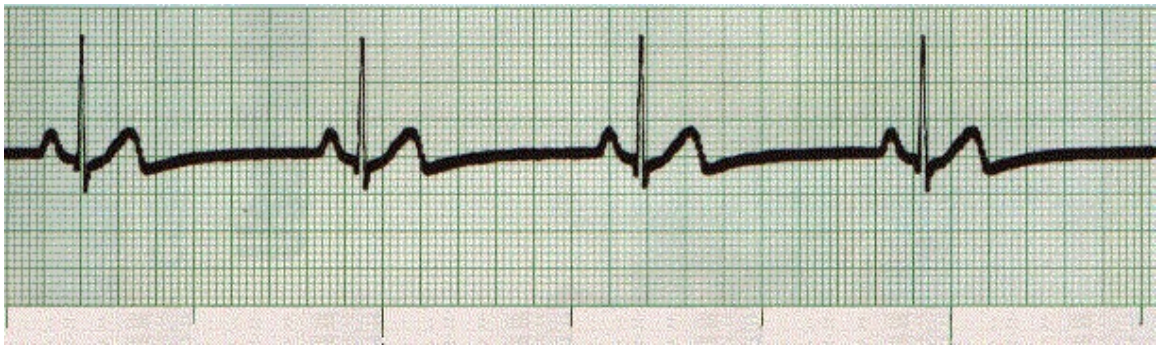
B. Ventricular fibrillation

C. Atrial fibrillation

D. Ventricular tachycardia

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PROBLEM 12



K.C. is a nineteen year old female who has come to the physician's office for a physical required to participate on the university swim team. She denies any past medical or surgical history and states that she swims laps for two hours almost every day. She denies any unusual symptoms or complaints and her physical exam is normal. How would you interpret this rhythm?

A. Normal sinus rhythm

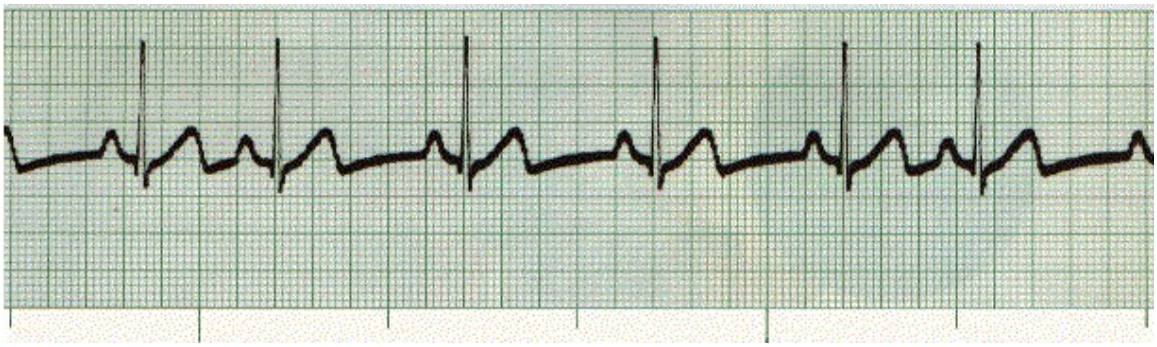
B. Normal sinus rhythm with PVCs

C. Junctional rhythm

D. Sinus bradycardia

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PROBLEM 13



F.D. is a sixty-one year old female who was admitted to the telemetry unit yesterday following a cardiac catheterization. She is scheduled for discharge this morning. Upon seeing this rhythm, you go to her room to check on her and you find her drinking a regular cup of coffee that her husband brought for her. You remind her that her doctor has strict orders against her drinking caffeinated beverages. She states that she feels just fine and has no symptoms at all. Her blood pressure is 124/62. How would you interpret this rhythm.

A. Sinus rhythm with PACs

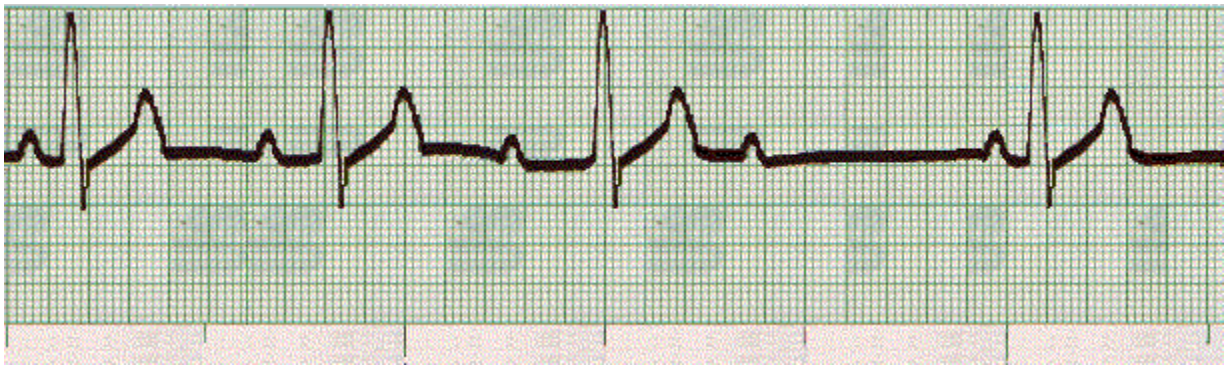
B. Sinus rhythm with PVCs

C. Atrial fibrillation

D. Normal sinus rhythm

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PROBLEM 14



R.K. is a 62 year-old male who has arrived in the emergency department complaining of dull, crushing, chest pain and shortness of breath. He is diaphoretic and pale. His cardiac enzymes are elevated and his twelve-lead ECG suggests an acute inferior myocardial infarction. How would you interpret this rhythm?

A. Third-degree AV block

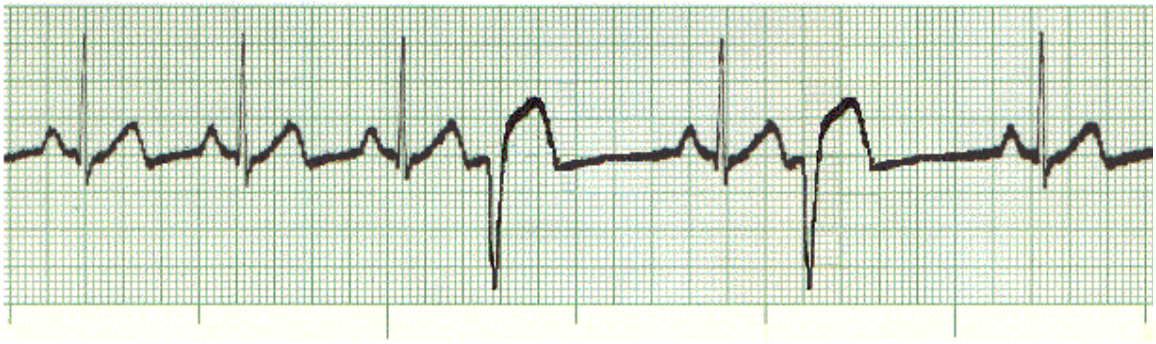
B. Second-degree AV block type I

C. Sinus bradycardia

D. Second-degree AV block type II

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PROBLEM 15



H.B. is a thirty-six year-old female who has arrived in the emergency department complaining of palpitations. She denies any shortness of breath, pain, or other symptoms. Her pulse is strong but irregular. Her blood pressure is 144/78. She has no past medical history. She admits to smoking two packs of cigarettes a day for the past fifteen years. How would you interpret this rhythm.

A. Normal sinus rhythm

B. Normal sinus rhythm with PACs

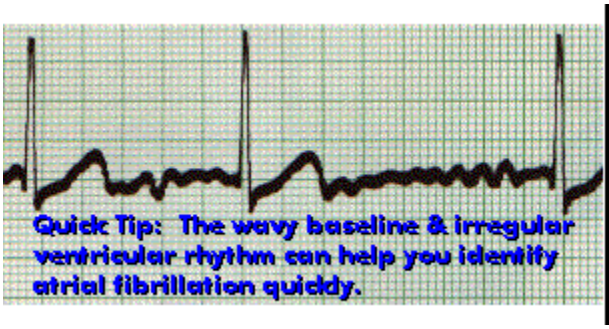
C. Normal sinus rhythm with PVCs

D. Sinus bradycardia

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ANSWERS

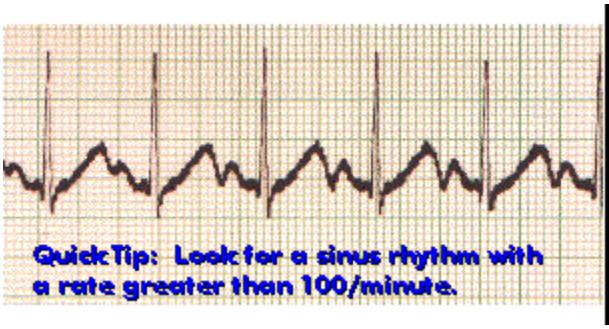
1. B



Atrial Fibrillation

Atrial Rate - up to 400/min
Ventricular Rate - 60-150
Atrial Rhythm - irregular
Ventricular Rhythm - iregular
P,QRS,T - all present & normal
PR Interval - inconsistent
QRS Interval - <0.12 secs

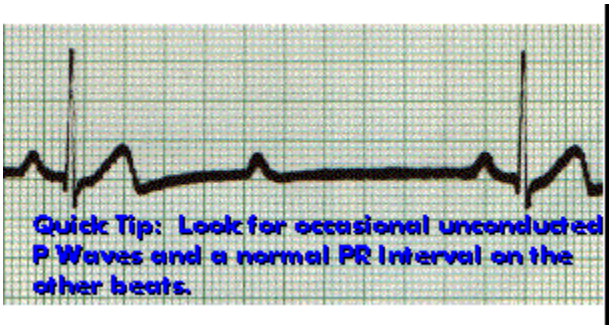
2. A



Sinus Tachycardia

Atrial Rate - >100/min
Ventricular Rate - >100/min
Atrial Rhythm - regular
Ventricular Rhythm - regular
P,QRS,T - all present & normal
PR Interval - <0.04 secs
QRS Interval - <0.12 secs

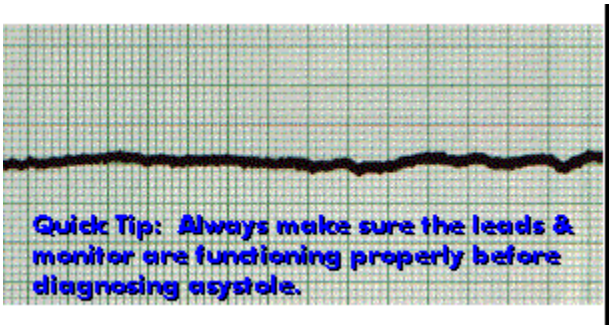
3. D



Second Degree Block Type 2

Atrial Rate - 60-100/min
Ventricular Rate - 20-100
Atrial Rhythm - regular
Ventricular Rhythm - regular
P,QRS,T - all present & normal
PR Interval - normal for conducted beats
QRS Interval - <0.12secs

4. B

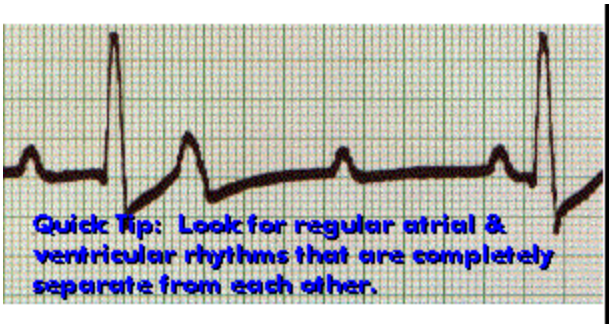


Asystole

Atrial Rate - N/A
Ventricular Rate - N/A
Atrial Rhythm - N/A
Ventricular Rhythm - N\A
P,QRS,T - none present
PR Interval - N/A
QRS Interval - N\A

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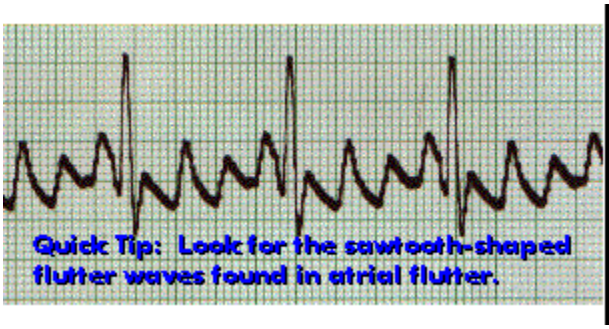
5. A



Third Degree Block

Atrial Rate - 60-100/min
Ventricular Rate - 20-40
Atrial Rhythm - regular
Ventricular Rhythm - regular
P,QRS,T - all present & normal
PR Interval - N/A
QRS Interval - >0.12 secs

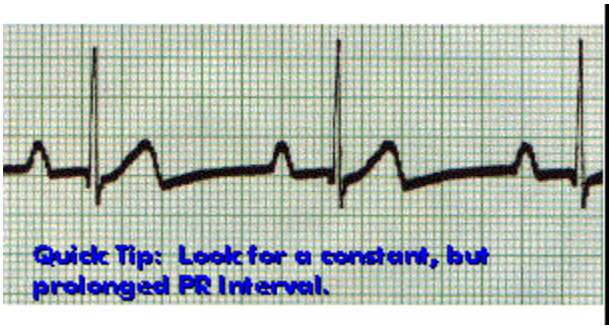
6. D



Atrial Flutter

Atrial Rate - 300-350/min
Ventricular Rate - variable
Atrial Rhythm - regular
Ventricular Rhythm - regular
P,QRS,T - flutter waves instead of P Waves
PR Interval - N/A
QRS Interval - <0.12 secs

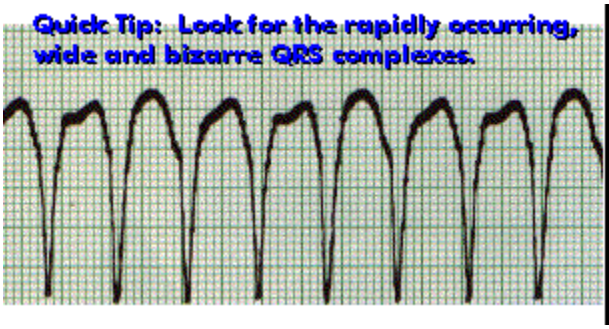
7. A



First Degree Block

Atrial Rate - 60-100/min
Ventricular Rate - 60-100
Atrial Rhythm - regular
Ventricular Rhythm - regular
P,QRS,T - all present & normal
PR Interval - >0.20 secs
QRS Interval - <0.12secs

8. C

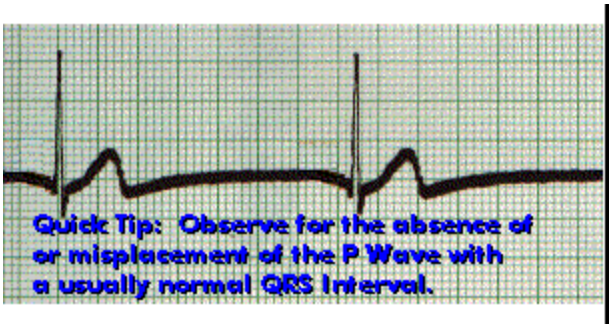


Ventricular Tachycardia

Atrial Rate - usually not seen
Ventricular Rate - 100-200
Atrial Rhythm - N/A
Ventricular Rhythm - regular
P,QRS,T - QRS & T present
PR Interval - N/A
QRS Interval - > 0.12 secs

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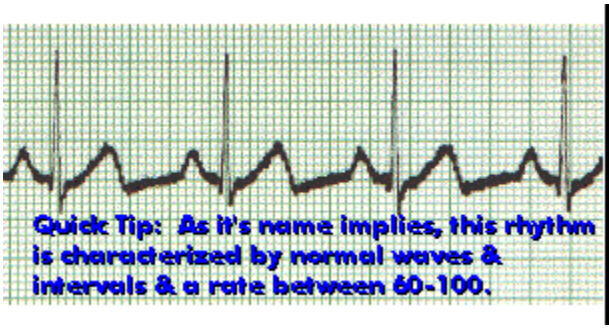
9. B



Junctional Rhythm

Atrial Rate - often not seen
Ventricular Rate - 40-60/min
Atrial Rhythm - often not seen
Ventricular Rhythm - regular
P,QRS,T - P Wave may be absent or inverted
PR Interval - N/A
QRS Interval - <0.12 secs

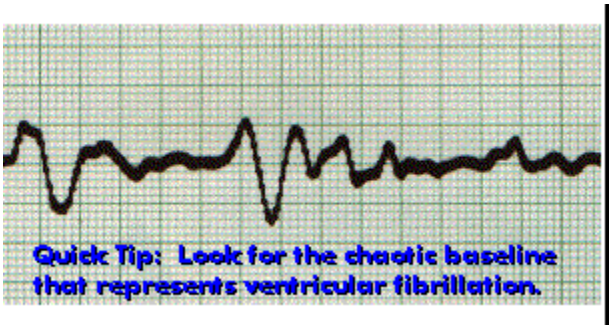
10. A



Normal Sinus Rhythm

Atrial Rate - 60-100/min
Ventricular Rate - 60-100/min
Atrial Rhythm - regular
Ventricular Rhythm - regular
P,QRS,T - all present & normal
PR Interval - <0.04 secs
QRS Interval - <0.12 secs

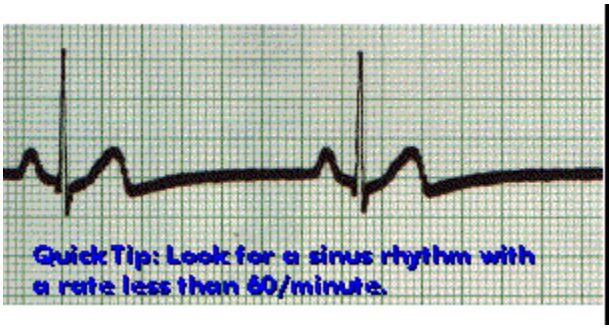
11. B



Ventricular Fibrillation

Atrial Rate - N/A
Ventricular Rate - N\A
Atrial Rhythm - N/A
Ventricular Rhythm - chaotic
P,QRS,T - indeterminate
PR Interval - N/A
QRS Interval - N/A

12. D

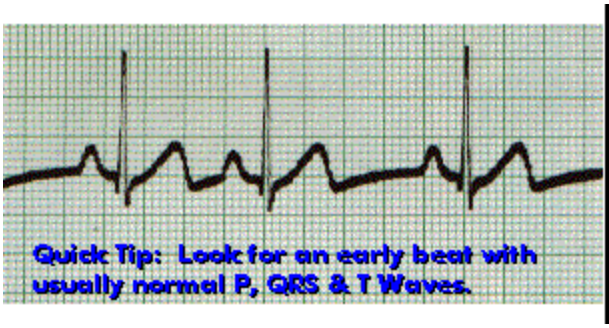


Sinus Bradycardia

Atrial Rate - <60/min
Ventricular Rate - <60/min
Atrial Rhythm - regular
Ventricular Rhythm - regular
P,QRS,T - all present & normal
PR Interval - <0.04 secs
QRS Interval - <0.12 secs

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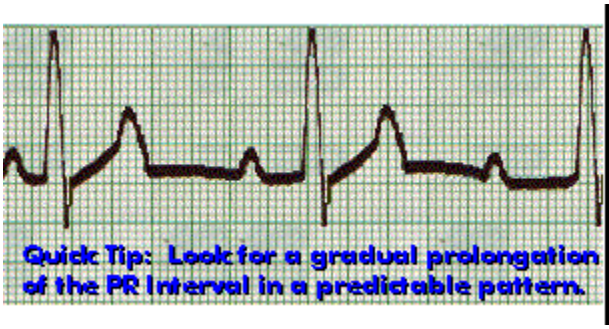
13. A



Premature Atrial Contractions

Atrial Rate - usually normal
Ventricular Rate - corresponds to atrial rate
Atrial Rhythm - irregular
Ventricular Rhythm - irregular
P,QRS,T - all present & normal
PR Interval - <0.20 secs
QRS Interval - <0.12 secs

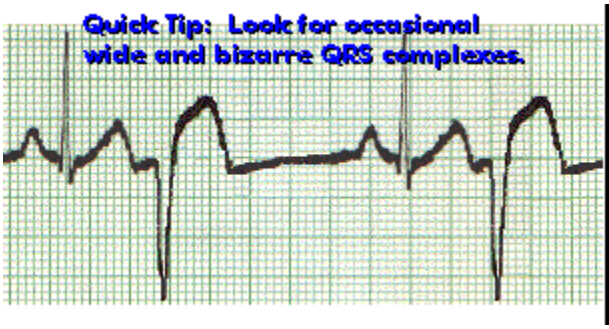
14. B



Second Degree Block Type 1

Atrial Rate - 60-100/min
Ventricular Rate - 60-100
Atrial Rhythm - regular
Ventricular Rhythm - irregular
P,QRS,T - all present & normal
PR Interval - prolongs in a gradual fashion until a beat is skipped
QRS Interval - <0.12secs

15. C



Premature Ventricular Contractions

Atrial/Ventricular Rate - corresponds to underlying rhythm
Atrial/Ventricular Rhythm - irregular during PVC
P,QRS,T - usually, only the QRS in visible in a PVC
QRS Interval - >0.12 secs during PVC

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